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Validation Diffraction Node in XMF

Analysis, Method and User Case Report

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Abstract

Validation Diffraction Node in XMF- Analysis, Method and User Case Report

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NOMENCLATURE

Symbol	Description	Unit
A	Amplitude	m
a	Added mass	kg
b_{lin}	Linear damping	Ns/m
b_{quad}	Quadratic damping	Ns^2/m^2
C_p	Pressure coefficient	$-$
Fr	Froude number	$-$
F_X	Force in x-direction	N
F_Y	Force in y-direction	N
F_Z	Force in z-direction	N
g	Gravitational acceleration	m/s^2
I	Turbulence intensity	$-$
k	Turbulence kinetic energy	m^2/s^2
L_{ref}	Reference length	m
p	Pressure	Pa
p_{ref}	Reference pressure	Pa
Re	Reynolds number	$-$
Sc	Scruton number	$-$
T	Draught	m
T_N	Natural period	s
T_{ref}	Reference period	s
u	Velocity	m/s
u_*	Friction velocity at the wall	m/s
U_{ref}	Reference velocity	m/s
y	Distance to the wall	m
y^+	Dimensionless distance to the wall	$-$
Δt	Time step	s
λ	Scale factor	$-$
ρ	Density	kg/m^3
μ	Dynamic viscosity	Pas
ν	Kinematic viscosity	m^2/s
ω	Turbulence dissipation per unit kinetic energy	m^2/s^3
ω	Vorticity	$1/s$
ζ	Damping ratio	$-$

Abbreviations:

CFD	Computational Fluid Dynamics
MARIN	Maritime Research Institute Netherlands
SI	Scientific International
VBM	Variational Boussinesq Model
DWF	Diffraction Wave Field Node

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1 INTRODUCTION

1.1 Background

Offshore vessels encounter waves in the field. These incoming waves are reflected by the vessels. The vessels themselves also generate or radiate waves as a result of their motions. MARIN has been developing hydrodynamic simulation software for many decades, and in the latest version of aNySIM (v2025.06.0), the time domain simulation program for analysis of multi-body dynamics in ships and offshore operations, it has implemented a new diffracted wave field node (DWF) that transform frequencies domain waves into time domain waves. aNySIM XMF is part of the Extensible Modelling Framework (XMF). The diffracted wave field node in XMF aims to compute the wave height and the water velocities at any location in the spatial domain, accounting for incoming, reflected and ultimately radiated waves. To transform frequency domain results into the time domain, it uses a combination of Impulse Response Theory, Interpolation, and the sum of interpolated functions. The frequency domain values are pre-processed, using a frequency domain diffraction analysis software, and stored in databases which then are read by XMF. MARIN has developed DIFFRAC, a frequency-domain potential-flow solver for radiation and diffraction loads. It uses a three-dimensional boundary-element method to solve the linearised velocity-potential problem and provides added mass, radiation damping, wave-excitation forces, and pressure fields. Irregular frequencies are mitigated using a rigid-lid formulation, and the method has been validated extensively in both shallow and deep water.
To compare and gain trust in this new node, the Variational Boussinesq Model – XMF (VBM) will be used. VBM is MARIN's nonlinear, depth-integrated wave solver based on a Hamiltonian Boussinesq formulation. It captures key dispersive and nonlinear effects while remaining computationally efficient. It's first purpose is to be used for training simulation in the simulator, however this wave model may also prove very valuable to perform multi-body "engineering" simulations for operational performance and safety assessments. Compared to the more classical multi-body database approach, the main advantage of the VBM model is that it allows the bodies to move with respect to each other at any time step without limitation. Other advantages also to be noted are: easy visualisation of the disturbed waves around the bodies and the possibility to include non-linear effect like breaking waves and propeller outflow. Validation studies were done for shallow and deep water, showing good approximation of the diffraction waves.

1.2 Objectives

In this research project, the main objective is to perform a sensitivity study with respect to the different parameters of the DWF, extract a simple user case advice for offshore vessels, and compare the accuracy of the method against VBM for diffraction and radiated waves.

1.3 Approach

To achieve the objective, the following steps are taken:

- Diffraction Frequency Analysis is performed with version 3.5.7b of DIFFRAC for all the study cases.
- Time-simulation in aNySIM with VBM and DWF is performed for all the study cases with version v2025.06.0.
- The effect of the space interpolation node in aNySIM (DWF) on the motions and wave elevation is investigated.
- The effect of the frequency interpolation node in aNySIM (DWF) on the motions and wave elevation is investigated.

- The effect of the angle interpolation node in aNySIM (DWF) on the motions and wave elevation is investigated.
- The effect of the vertical step size in DIFFRAC on the motions and wave elevation is investigated.
- The Sensibility study with respect to the temporal and spatial grid on the the motions and wave elevation is investigated.

To investigate the results, we will make use of Rayleigh statistics and standard data analysis tools. This approach will allow us to study the results at a statistical level, rather than a deterministic one, making them more reliable.

1.4 The Problem

The case study used in this report consists of on a single-body LNG carrier, represented by MARIN's m8383 model (Figure 1.1). Given that one of the primary application areas for this study is the offshore installation

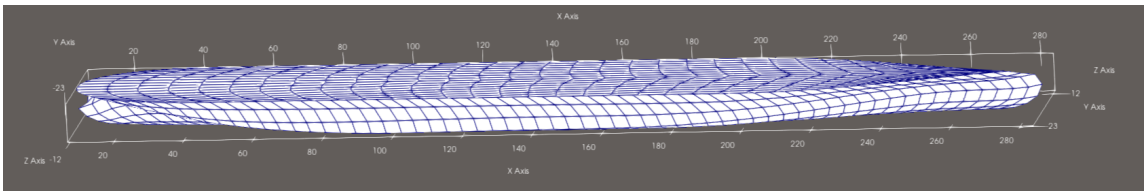

Figure 1.1: MARIN m8383 LNG carrier model with 1500 panels.

of wind turbines, particular attention is paid to the prediction of the diffraction wave field generated by the vessel's shielding effect. For this reason, the analysis focuses on the vessel's port side, where these effects are most pronounced.
To support flexibility and reproducibility, the computational framework developed for this study—comprising both the code repository and the associated Python library—has been designed in a modular manner. This allows for straightforward adaptation of the analysis domain, making it possible to extend the study to consider wave fields around the entire vessel or to restrict the evaluation to specific regions of interest, such as the starboard side, as well as change the vessel type and geometry. This design choice ensures that the methodology can be easily reused and adapted for different configurations and application scenarios.

To further analyse the spatial variation of the diffraction wave field, the vessel domain is subdivided using five longitudinal reference lines. These lines are defined relative to the hydrodynamic centre of the vessel, which is taken as the primary reference point. The five positions correspond to:

$$-2x_{\text{HydCenter}}, \quad -\tfrac{1}{2}x_{\text{HydCenter}}, \quad x_{\text{HydCenter}}, \quad +\tfrac{1}{2}x_{\text{HydCenter}}, \quad +2x_{\text{HydCenter}}.$$

These locations are determined with respect to the hydrodynamic centre, as illustrated schematically in Figure 1.2. In the present study, the longitudinal direction of the vessel is denoted as the *position* (i.e., the x-axis), while the transverse direction away from the vessel is referred to as the *distance from the vessel* or *depth* (i.e., the y-axis).
Wave probes, as implemented in aNySim, are used to extract the free-surface elevation and diffraction field around the structure. These probes are aligned along the five reference lines described previously. Starting from the central line of the vessel, the probes are positioned at distances of 1 m, 2.5 m, and 5 m. For studies involving spatial interpolation, the probe locations are systematically shifted by 1 m in the transverse direction to assess sensitivity and reconstruction accuracy.

The maximum transverse extent of the probe domain is set to 50 m from the vessel centreline. For reference, the vessel beam is 44.2 m, corresponding to a half beam of approximately 22.1 m. This ensures that the probe field extends well beyond the immediate vicinity of the hull, capturing both near-field and intermediate diffraction effects.

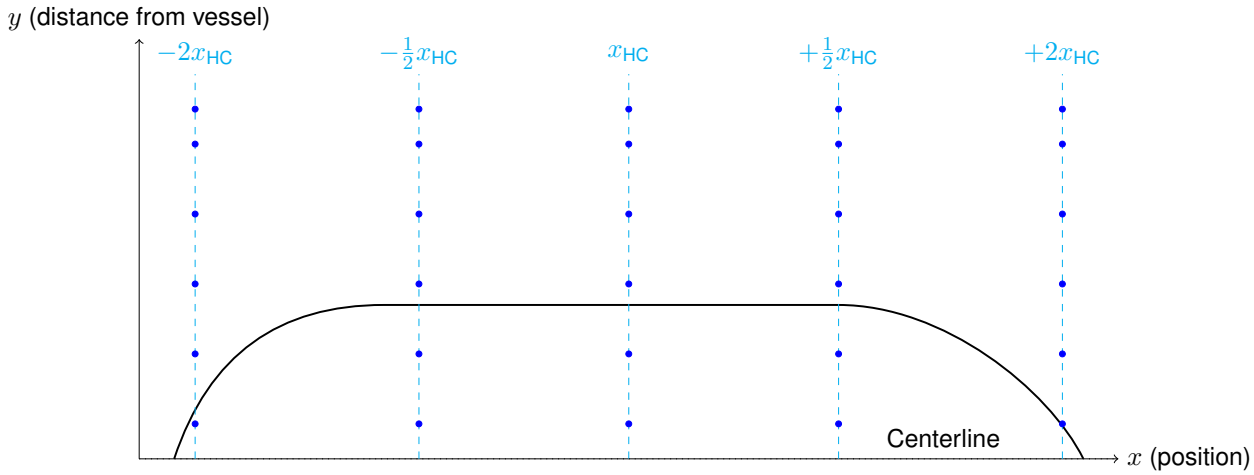

Figure 1.2: Schematic representation of the half vessel and the five longitudinal reference lines defined relative to the hydrodynamic centre (x_{HC}). The extreme positions $-2x_{HC}$ and $+2x_{HC}$ are located near the aft and bow regions, respectively, while the intermediate lines lie within the vessel length. Wave probes (shown for the 5 m spacing configuration only) are aligned along these lines and extend up to 50 m from the vessel centreline.

In addition to the spatial discretisation, a range of incident wave directions is considered in order to fully characterise the diffraction behaviour. The wave direction β is varied from 30° to 150°, covering oblique to beam-sea conditions. Depending on the level of resolution required for a given study, the angular discretisation is chosen as either 15° or 30° increments.

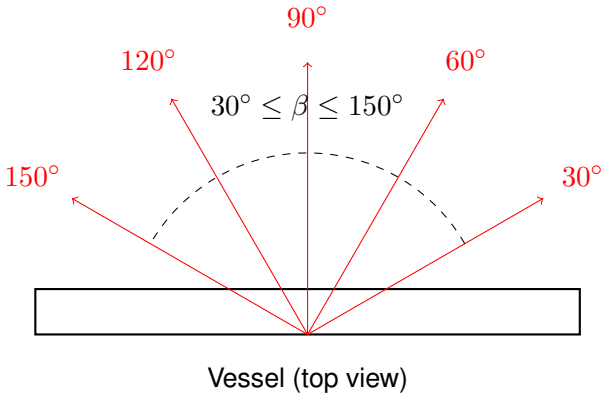

Figure 1.3: Wave incidence directions considered in the study. The wave angle β ranges from 30° to 150°, with increments of 15° or 30° depending on the analysis.

Finally, it should be noted that the present analysis is restricted to regular (monochromatic) wave conditions. All simulations are performed using incident waves with a constant amplitude of 1 m and angular frequencies between $0.05\mathrm{rad/s}$ and $2\mathrm{rad/s}$. This choice provides a controlled and consistent framework for evaluating the diffraction effects and facilitates direct comparison between different configurations, probe locations, and wave directions. The use of regular waves allows for a clear interpretation of the vessel-induced wave field without the additional complexity introduced by irregular sea states.

1.5 Content of the report

The report is structured as follows. It begins with a brief chapter introducing the theoretical background and the main aspects of the numerical implementation. This section provides the fundamental concepts required to understand the modelling approach and the computational framework adopted in this study.
This is followed by the section on *Numerical Setup and Verification*. In this part, an initial benchmark problem is defined and implemented across all the software tools employed. The general structure of the computational repository is described, highlighting its modular design and flexibility. Verification is then carried out to ensure that the obtained results are consistent, physically meaningful, and comparable across the different solvers.
The next section focuses on *Diffracted Waves*, where a detailed sensitivity analysis is performed. The influence of several numerical and discretisation parameters is investigated, including the number of panels used for the vessel representation, the time step, the spatial resolution, the vertical discretisation, and the treatment of angular and frequency interpolation. Additionally, the effect of spatial interpolation is analysed. The results are systematically compared, and conclusions are drawn, including a comparison against reference VBM (validated benchmark model) results.
A similar methodology is then applied in the subsequent section on *Diffracted and Radiated Waves*. Here, the combined effects are analysed, and the sensitivity studies are extended to include both wave diffraction and radiation phenomena, following the same set of evaluation criteria as in the previous section.
Finally, the report concludes with a summary of the main findings and key observations derived from the study, together with recommendations for future work.
Additional theoretical background, extended documentation, and supplementary results are provided in the appendices.

2 Theoretical Backgrounds and Implementation

In this section, we briefly review the theoretical background underlying the numerical methods considered in this study. Further details can be found in the cited literature as well as in internal documentation.
A basic understanding of Ocean Surface Waves, Rigid Body Dynamics, Potential Theory, Diffraction Theory in the Frequency Domain, and vertically averaged methods—particularly the Variational Boussinesq Model—is assumed. Concise introductions to these topics, along with relevant references, are provided in the Appendix.
We begin by outlining the theoretical differences between the two numerical approaches employed in this work: (i) diffraction theory in the frequency domain, combined with a Fourier transform to obtain time-domain responses, and (ii) the Variational Boussinesq model. Subsequently, we briefly describe their respective numerical implementations and discuss the primary limitations associated with each method.

2.1 Diffraction Theory vs Variational Boussinesq Model

2.2 Diffraction Wave Field Node (DWF)

2.2.1 Node Implementation Overview

A clear understanding of the diffraction node implementation is essential prior to validating any post-processed results. According to the *DIFFRACTION Library* documentation (v2025.06.0), the node does not directly utilize the incoming wave components. Instead, it operates based on an internally defined wave field.
When the characteristics of the incoming waves—such as direction, frequency, or spatial location—do not exactly coincide with the entries available in the internal database, the node performs interpolation between the closest data points. This interpolation procedure is consistently applied across all relevant dimensions (directional, spectral, and spatial). However, the documentation does not specify the interpolation scheme or its order, which introduces uncertainty regarding the smoothness and accuracy of the reconstructed wave field.
The interpolated solution is then superimposed onto both the current state and the wave field. This procedure may introduce numerical artifacts, particularly in regions where the database resolution is limited or where strong gradients are present. Such effects must therefore be carefully considered during the verification process.

2.2.2 Wave Excitation

Wave excitation describes the interaction between incident waves and the vessel. In this framework, the incoming wave field is represented as a linear superposition of spectral components derived from linear wave theory.
The associated hydrodynamic loads are obtained from a pre-computed or experimentally measured database, which provides force and moment coefficients for a range of wave frequencies and propagation directions.
In the time domain, the total excitation is evaluated by summing the contributions of all spectral components.
As a result, the accuracy of the excitation forces depends directly on:

- The resolution and fidelity of the wave spectrum (in both frequency and direction),
- The completeness and quality of the hydrodynamic database.

First-Order Wave Excitation The first-order response of the vessel to monochromatic wave components is described by the Response Amplitude Operator (RAO), which relates the wave amplitude to the resulting forces and moments. The excitation load vector $\mathbf{q}$ is given by:

$$q_k = \sum_{i=1}^{n} \mathrm{Re}\left[\mathrm{RAO}_k(\omega_i, \mu_i)\, \bar{a}_i\right], \quad k = 1, \ldots, 6 \tag{1}$$

where ω_i and μ_i denote the frequency and direction of the i-th component, and a_i is the corresponding complex wave amplitude.

Second-Order Wave Excitation Second-order effects account for nonlinear interactions between pairs of wave components. These interactions are described by the Quadratic Transfer Function (QTF), leading to the following expression:

$$q_k = \sum_{i=1}^{n} \sum_{j=1}^{n} \mathrm{Re}\left[\mathrm{QTF}_k(\omega_i, \omega_j, \mu_i, \mu_j)\, \bar{a}_i a_j\right], \quad k = 1, \ldots, 6 \tag{2}$$

Due to its four-dimensional nature, the QTF database can become computationally demanding. In practical applications, several simplifications are typically introduced:

- Neglecting high-frequency interaction terms ($\omega_i + \omega_j$),
- Restricting interactions to a limited frequency difference band,
- Omitting directional coupling between components.

2.2.3 Wave Radiation

Wave radiation loads arise from the motion of the vessel itself and are derived from hydrodynamic databases generated through frequency-domain diffraction analysis. These databases provide added mass $\alpha(\omega)$ and damping $\beta(\omega)$ coefficients.
To obtain time-domain radiation forces, impulse response theory is employed. The resulting load is expressed as a convolution integral:

$$y(t) = \int_0^{\infty} v(t-\tau)\, k(\tau)\, d\tau \tag{3}$$

where $v(t)$ denotes the velocity and $k(\tau)$ is the impulse response function.
In practice, $k(\tau)$ is computed from the damping coefficients as:

$$k(\tau) = \frac{2}{\pi} \int_0^{\infty} \beta(\omega) \cos(\omega\tau)\, d\omega \tag{4}$$

Since this formulation does not capture the frequency-independent contribution of the added mass, a correction term involving α_∞ must be introduced separately.

2.2.4 Definition of the Vertical Coordinate

The vertical coordinate z is defined relative to the free surface, with $z = 0$ corresponding to the undisturbed water level. Negative values indicate positions below the free surface.
The parameters $z_{\min}$ and $z_{\max}$ specify the vertical extent of the computational grid exported from the diffraction solver. These bounds define the region over which the wave field is physically represented and numerically valid.

2.2.5 Module Evaluation Behavior

The evaluation of the diffraction module follows a set of predefined rules:

- For points located outside the (x, y) grid or below $z_{\min}$, the contribution is set to zero.
- For points above $z_{\max}$ (i.e., above the free surface), the solution is obtained through interpolation using $z_{\max}$, effectively projecting the free-surface results upward.
- For input parameters outside the available database range (in frequency or direction), the behavior is not defined.

This latter aspect represents a potential source of uncertainty and must be accounted for during result interpretation.

2.2.6 Limitations

The principal limitations of the present formulation can be summarized as follows:

- The interpolation scheme within the diffraction node is undocumented, leading to uncertainty in accuracy and smoothness,
- Strong dependence on the resolution and quality of the hydrodynamic databases,
- Significant computational cost associated with second-order wave excitation,
- Assumptions of linear wave radiation and small-amplitude vessel motions,
- Undefined behavior outside the parameter range covered by the database.

2.3 Variational Boussinesq Model (VBM)

2.3.1 Implementation Overview

2.3.2 Wave Field

The beach Zones

2.3.3 Vessel Pressure Field

The Variational Boussinesq Model (VBM) accounts for the pressure exerted by the vessel on the wave field through a dedicated projection method. This approach introduces a localized deformation of the free surface. To quantify the spatial extent of this influence, a statistical analysis was performed using 91 solutions from a captive vessel case.
Specifically, we evaluated the location at which the mean wave elevation returns to approximately zero, indicating a negligible effect of the vessel-induced pressure. As expected, the variability of this metric across the dataset is minimal. Considering that the vessel has a maximum beam of $44.2\,\mathrm{m}$ (i.e., $22.1\,\mathrm{m}$ from the centerline), the results indicate that the pressure-induced deformation of the wave field persists up to approximately 1–$2\,\mathrm{m}$ away from the vessel boundary.
It should be noted that no additional analysis was conducted to assess whether placing wave probes outside the vessel nodes alters this observation.

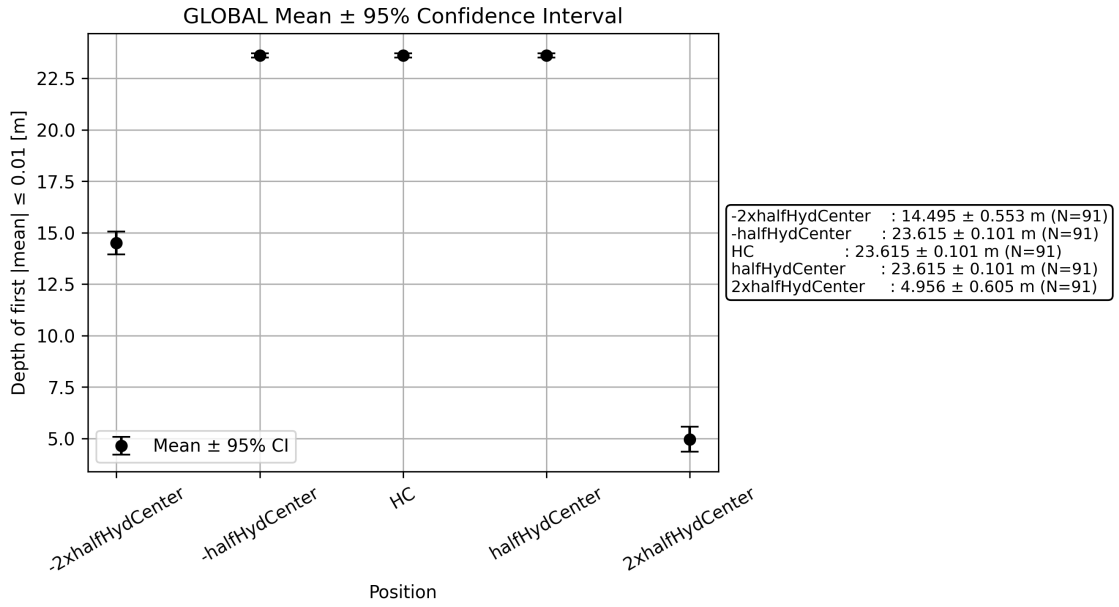

Figure 2.4: Confidence interval analysis of the transverse location (denoted as depth) at which the mean wave amplitude approaches zero, indicating negligible influence of vessel-induced pressure on the wave field. The position is defined along the x-axis with respect to the hydrodynamic center. See Introduction: The Problem.

2.3.4 Limitation

The VBM presents several inherent limitations that must be considered when interpreting the results:

- Numerical stability and convergence are sensitive to discretization choices, particularly for shorter wave conditions;
- Absorbing (beach) zones must be sufficiently large to ensure full wave absorption, especially for oblique wave incidence; otherwise reflections may affect the solution;
- The spatial discretization needs to be sufficiently small to ensure that the grid around the ship has sufficient cells for the CUDA solver to work;
- The VBM needs a cool-off period to converge to the result;
- The VBM pressure vessel component deforms the wave field around it, meaning it creates a band region around the vessel where the wave probe solution is influenced by it.

3 Numerical Setting

to be done

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4 Diffraction Waves

4.1 Introduction

In this section, the vessel is considered captive under incoming wave conditions. The wave field obtained from the Diffraction Wave Field (DWF) node is compared against the superposition of incident and diffracted waves computed using the Variational Boussinesq Model (VBM).
The analysis begins with a sensitivity study focusing on the temporal and spatial discretization of the DWF node. In particular, the influence of the time step and grid resolution is investigated. Wave probes are placed directly at the (x, y) grid nodes and in the vertical domain positions described by the z-levels used in the DIFFRAC computations, ensuring a consistent comparison between both approaches.

4.2 Sensitivity Study

A sensitivity study is performed to establish practical guidelines for selecting the time step and spatial resolution. In addition to the horizontal grid, particular attention is given to the vertical spacing between the grids exported in the `.h5m` database from DIFFRAC, as this may affect the interpolation accuracy and the representation of the wave field.

4.2.1 Influence of Geometry Discretization

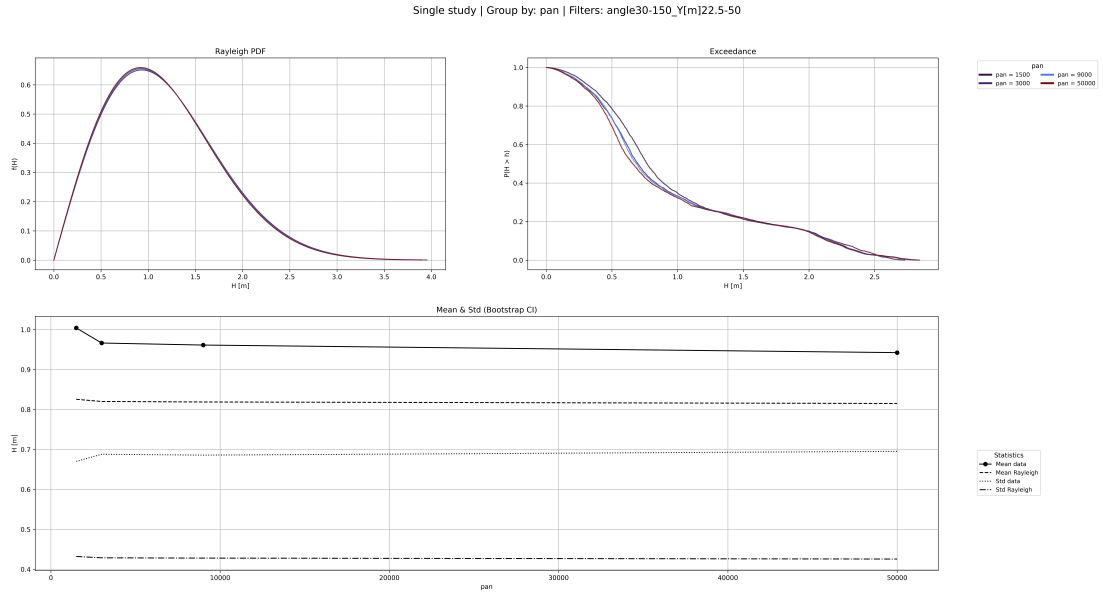
To reduce the overall computational cost of the sensitivity analyses, a preliminary investigation was carried out to assess whether the number of panels used in the geometric discretization of the vessel significantly influences the quality of the solution.

A comprehensive set of simulations was conducted, covering all incident wave directions, spatial resolutions ($\Delta x = 2.5$ m and 5 m), wave frequencies, and varying temporal discretizations expressed in points per wave period. For each configuration, the resulting free-surface elevation time series were analysed statistically.
As shown in Fig. 4.5, the wave height distributions were compared to the theoretical Rayleigh distribution, which is expected for narrow-banded Gaussian wave processes. The fitted Rayleigh probability density functions exhibit excellent agreement[1] with the empirical distributions across all considered sample sizes ($N = 1500$, 3000, 9000, and $50\,000$). Both the peak location and the overall shape are accurately reproduced, indicating that the simulated wave fields capture the expected second-order statistics.

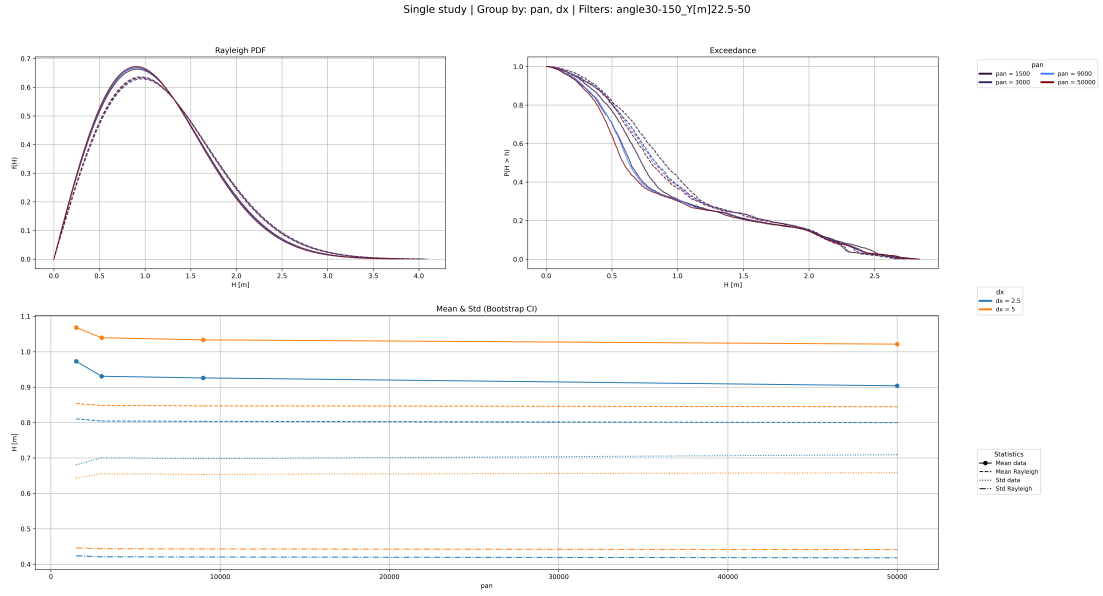
Minor discrepancies are observed for the smallest dataset ($N = 1500$), where a slight overestimation appears in the intermediate amplitude range. However, when analysing the exceedance probability—defined as the probability that the wave amplitude exceeds a given threshold—all four cases exhibit nearly identical tail behaviour. This confirms that extreme values are consistently represented even for relatively small samples.

As the sample size increases, the discrepancies rapidly diminish. For $N \geq 9000$, the agreement with the theoretical exponential decay becomes very strong, including in the tail region, which is critical for extreme-value analysis. Increasing the sample size therefore primarily reduces statistical noise without altering the underlying distribution.

[1] The observations of Longuet-Higgins (1980) and Holthuijsen (2007) indicate that the theoretically estimated significant wave height from the spectrum should be reduced by approximately 5%–10% in order to properly represent the real value due to linearity assumptions.



(a) Comparison of empirical wave height distributions with the fitted Rayleigh distribution for different sample sizes ($N = 1500, 3000, 9000,$ and $50\,000$).



(b) Same comparison including the effect of spatial resolution ($\Delta x = 2.5$ m and 5 m).

Figure 4.5: Statistical comparison of simulated wave heights against the theoretical Rayleigh distribution for different discretization levels.

Overall, the results indicate that the simulated wave heights are well described by Rayleigh statistics. No significant differences are observed beyond $N = 3000$, suggesting that statistical convergence with respect to the geometric discretization is effectively achieved at this resolution. Based on these findings, a discretization of 3000 panels is adopted for all subsequent simulations.

4.2.2 Time Step Sensitivity Study

A systematic sensitivity study was conducted to quantify the influence of temporal discretization on the accuracy of the simulated wave field. The analysis considers incident wave angles ranging from 30° to 150°, combined with a broad set of time steps defined as fractions of the wave period, namely $\Delta t = T/10, T/50$, $T/100$, $T/200$, $T/400$, and $T/800$. In addition to the temporal resolution, the study also includes three spatial grid resolutions ($\Delta x = 1$ m, 2.5 m, and 5 m), enabling a comprehensive assessment of the coupled effects of time step, spatial discretization, wave frequency, and probe location. The spatial coordinates are described along the y-direction[2] and along the x-direction[3].

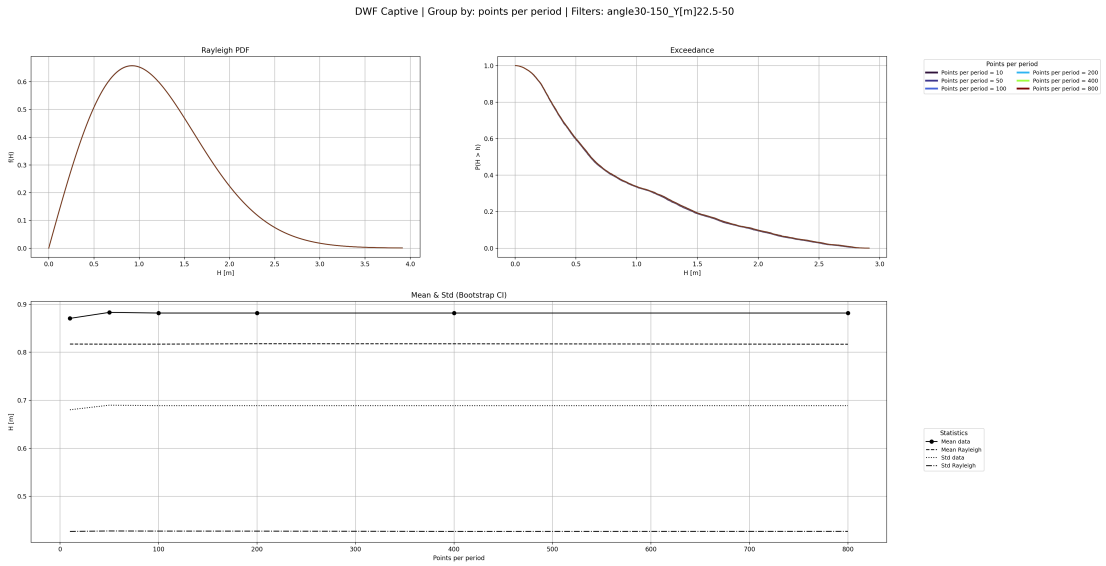


Figure 4.6: Statistical Analysis of the wave height (H) for different time steps (Δt) , showing convergence toward a stable distribution as temporal resolution increases.

Influence of Time Step Figure 4.6 presents the probability density functions (PDFs) of the wave height (H) for different time steps, aggregated over all tested configurations. It can be observed that the PDFs collapse onto a single curve as the time step is refined, indicating weak sensitivity to temporal resolution beyond the $T/100$ threshold. A more detailed view is provided in Figure 4.7, where the dependence on wave frequency is shown. The convergence behavior remains consistent throughout the frequency range, with negligible differences for $\Delta t \leq T/100$.

Figures 4.8 and 4.9 illustrate the influence of probe location (depth) and wave direction, respectively. In both cases, the effect of the time step remains secondary compared to physical variations induced by these parameters.

The spatial variation along the vessel axis is shown in Figure 4.10. While the statistical properties change with position due to physical effects, the convergence with respect to Δt remains robust. A slightly slower convergence is observed near the extremities of the domain compared to the central region.

In Figure 4.11 we have compare the wave height with respect to the x-direction for the ranges of angles 30-90 degrees and 90-150 degrees. From this study we can see that biggest difference is located on the same x-half plane of the incident wave. This because on the extreme x location of that plane is were we are going to have the most disturbed wave field.

[2] referred sometimes to as “depth”, i.e., distance from the vessel centerline

[3] referred to as “position”, parallel to the vessel axis

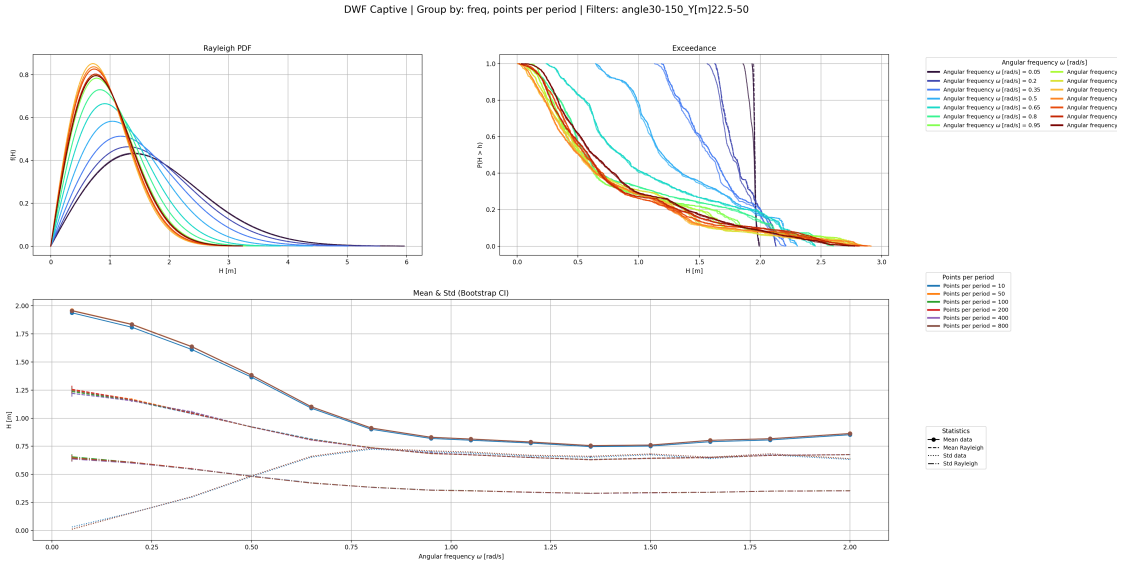


Figure 4.7: Statistical Analysis of the wave height (H) distributions across different frequencies. Convergence is achieved for $\Delta t \leq T/100$ with minimal variation.

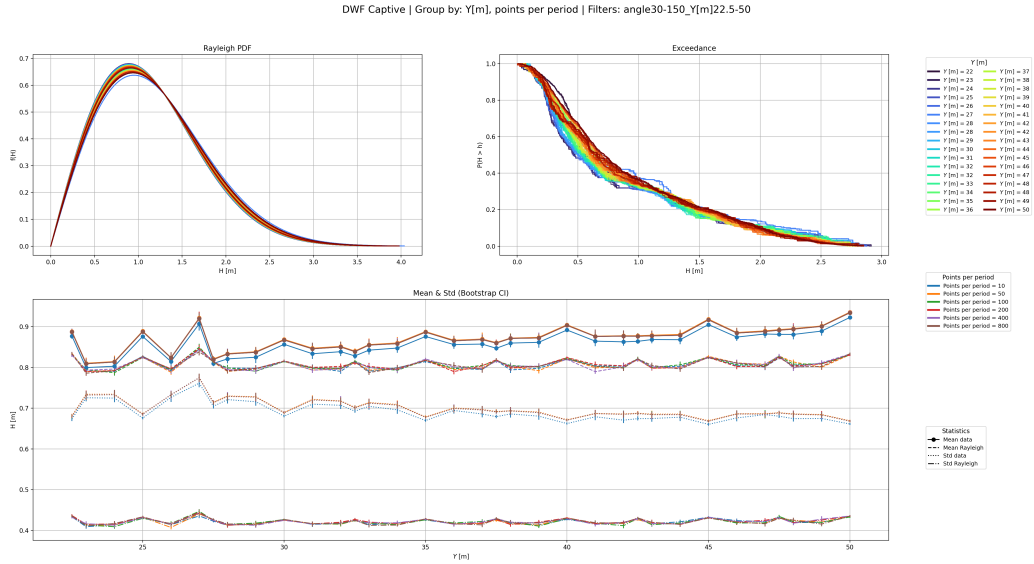


Figure 4.8: Dependence of the wave height distribution on probe depth for different time steps. Temporal discretization has a negligible influence compared to spatial variation.

Overall, the results clearly demonstrate that the influence of the time step becomes negligible once a moderate temporal resolution is achieved. Only the coarsest time step ($\Delta t = T/10$) exhibits noticeable deviations, reflected in slight distortions of the distributions and exceedance probabilities. These discrepancies rapidly vanish as the time step is reduced.

Combined Effect with Spatial Resolution The interaction between temporal and spatial discretization is illustrated in Figure 4.12, where results for different grid sizes are compared. While spatial resolution has a measurable impact—particularly in the tails of the distributions—the effect of time step remains consistent across all grids.

These results indicate that temporal and spatial discretization errors are largely decoupled within the inves-

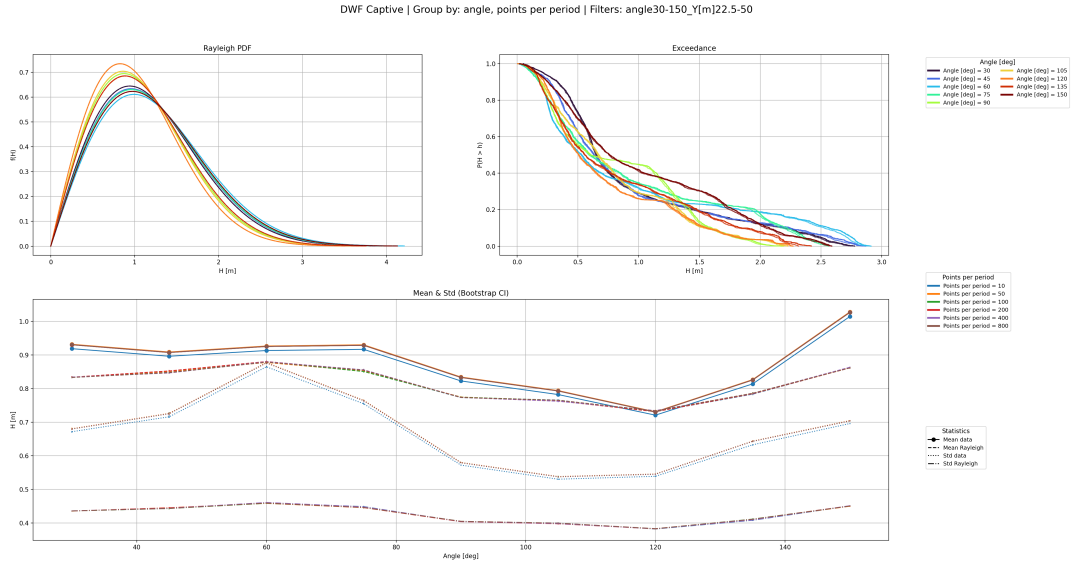


Figure 4.9: Effect of wave incidence angle on wave height distributions for varying time steps. Convergence trends remain consistent across all directions.The results indicate that the numerical scheme is robust with respect to directional wave propagation.

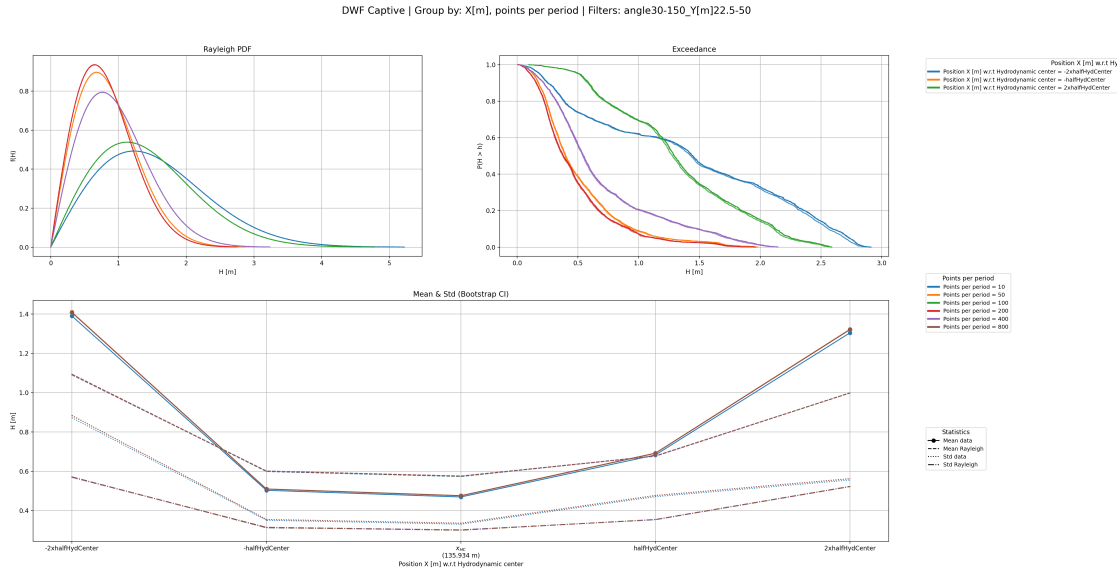
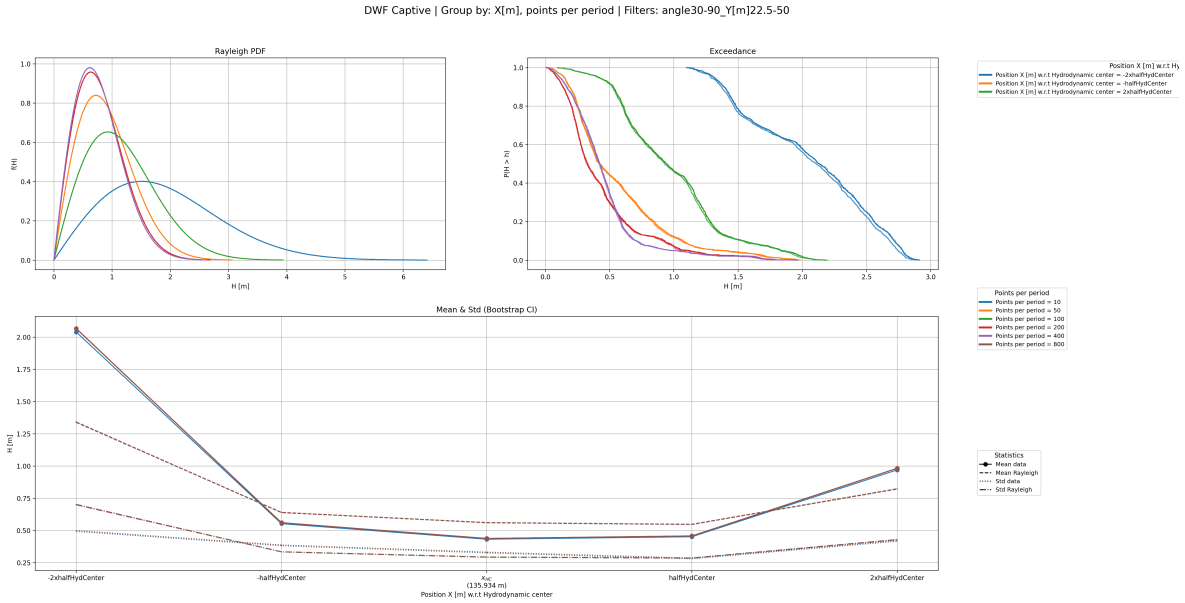


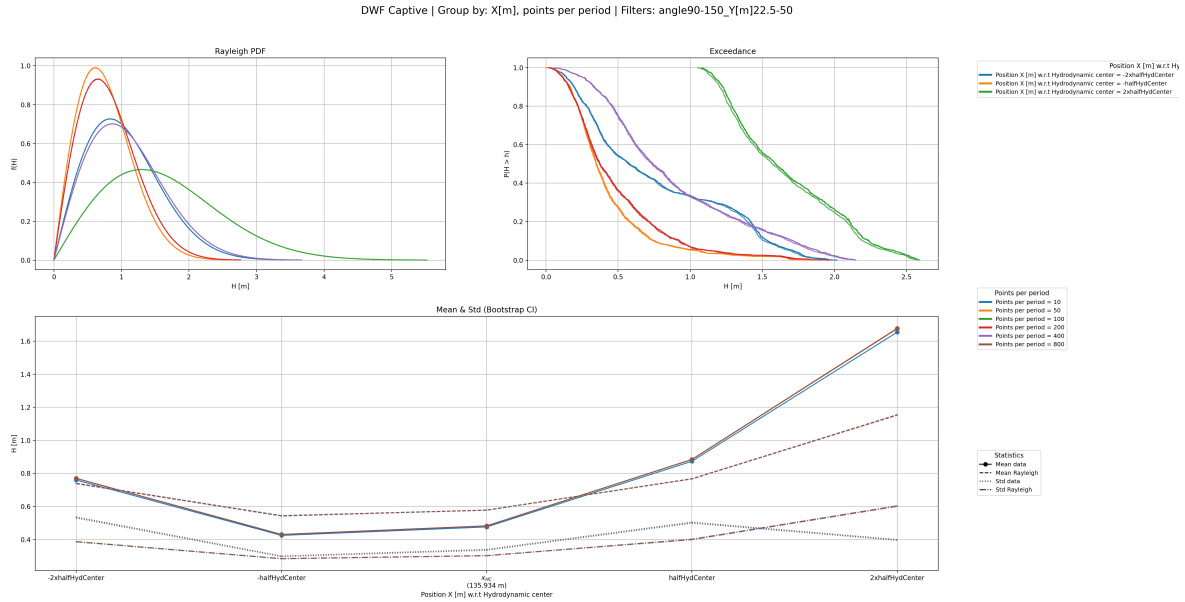
Figure 4.10: Wave height distributions along the x-direction (vessel axis) for different time steps. Convergence is slightly slower near the domain boundaries.

tigated parameter space. In particular, increasing the number of points per wavelength (spatial resolution) has a more pronounced effect on accuracy than increasing the number of time steps per period. This is in line with the theory behind the theory implemented in the diffraction analysis tools, `DIFFRAC` .

Summary The sensitivity analysis demonstrates that the numerical solution is only weakly dependent on the time step once a moderate temporal resolution is reached. A time step of $\Delta t \approx T/100$ is sufficient to ensure converged and reliable results across all investigated conditions, including variations in grid size, wave frequency, spatial location, and wave direction. Further refinement ($T/200$ and smaller) does not yield significant improvements in accuracy but increases computational cost.



(a) Wave height statistics with respect to X location and points per period for angle 30-90.



(b) Wave height statistics with respect to X location and points per period for angle 90-150

Figure 4.11: Statistical comparison of simulated wave heights along the X position between angles 30-90 and 90-150.

Additionally, the results indicate that convergence is slightly faster in regions close to the vessel center, while a slower convergence rate is observed near the extremities of the computational domain, in particular in the corresponding x-half plane of the incident wave. Meaning, studies that needs accurate results on the same half plane of the incoming waves, using a finer refinement in time discretization is advisable.

With this study, we can claim that choosing a time step corresponding to around 50-100 points per wave period (ppp) brings convergence results, although even a 10 ppp case has a difference of less then 0.05m in wave height, meaning a difference of at most 5%.

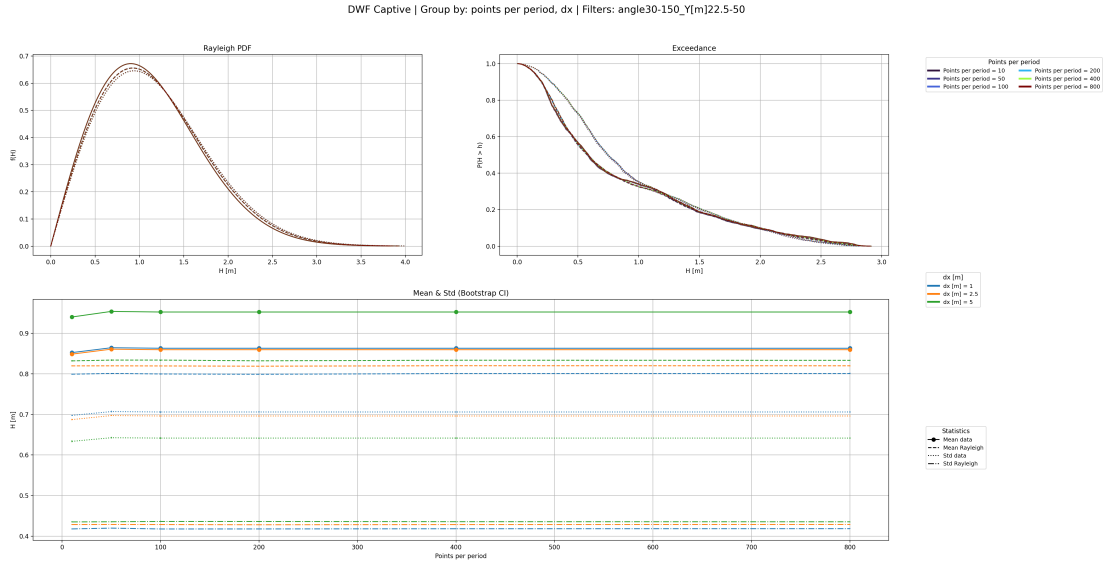


Figure 4.12: Combined influence of time step and spatial resolution on the wave height distribution. Spatial discretization has a stronger effect than temporal discretization beyond moderate resolution.

4.2.3 Spatial Step Sensitivity Study

A detailed sensitivity analysis was conducted to quantify the influence of spatial discretization on the simulated wave field. Three grid resolutions were considered, namely $\Delta x = 1$ m, 2.5 m, and 5 m. All other numerical parameters were kept consistent, including a sufficiently refined temporal discretization ($\Delta t \leq T/100$), ensuring that temporal errors remain negligible and do not interfere with the spatial convergence assessment.

The analysis covers a wide range of conditions, including wave incidence angles between $30°$ and $150°$, angular frequencies spanning $\omega = 0.05$–2 rad/s, and probe locations distributed along both the longitudinal (x) and transverse (y) directions.

Global Influence of Spatial Resolution The global impact of spatial resolution on the wave amplitude distribution is illustrated in Figure 4.13. The probability density functions (PDFs) obtained for the different grid sizes exhibit a high degree of consistency, with only minor deviations observed for the coarsest resolution. In particular, the $\Delta x = 5$ m case produces a slightly smoother distribution, reflecting a reduced ability to resolve steep gradients and extreme values.

This behaviour is further confirmed by the exceedance probability curves, which remain largely unchanged across resolutions, indicating that the global statistical characteristics of the wave field are well preserved. The maximum discrepancy between the $\Delta x = 2.5$ m and 5 m grids is approximately 0.1 m in wave height (equivalent to 0.05 m in amplitude), corresponding to about 10% in wave height.

A direct comparison between $\Delta x = 5$ m and $\Delta x = 2.5$ m is shown in Figure 4.14. The results demonstrate a near-perfect overlap of the Rayleigh distributions and exceedance curves, confirming that the overall statistical behaviour is robust with respect to spatial resolution. Nevertheless, a systematic bias is observed in the mean wave height, with the coarser grid exhibiting deviations on the order of 8–12%. This indicates that while the global distribution is preserved, small but consistent differences remain in the first-order statistics.

Dependence on Wave Direction The dependence of spatial sensitivity on wave incidence angle is presented in Figure 4.15. While directional effects significantly influence the wave field, the relative impact of spatial discretization remains moderate across all directions.

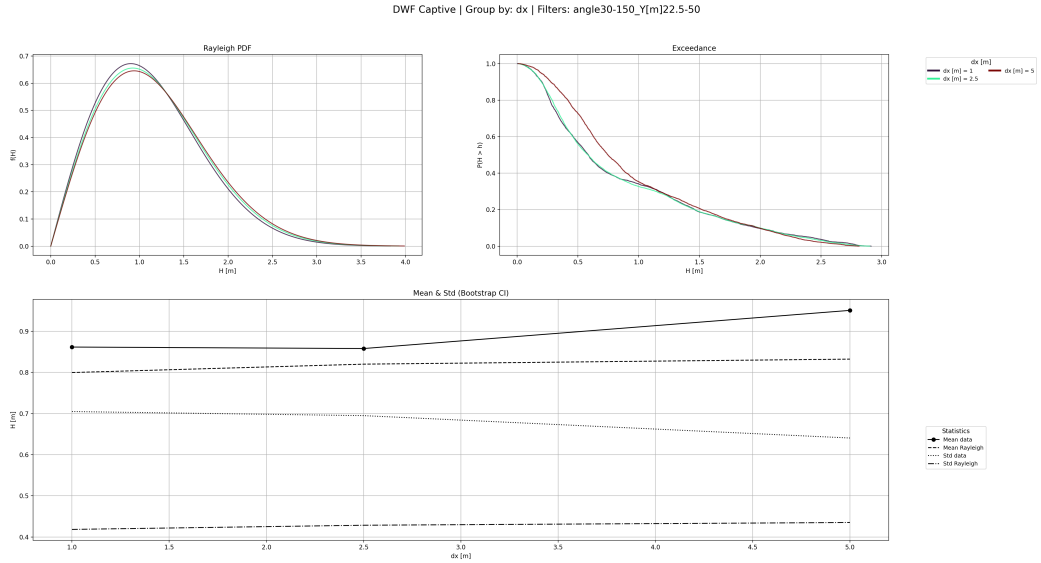


Figure 4.13: Global comparison of wave height PDFs for different spatial resolutions. The distributions are largely consistent, with slight smoothing effects observed for the coarsest grid ($\Delta x = 5$ m).

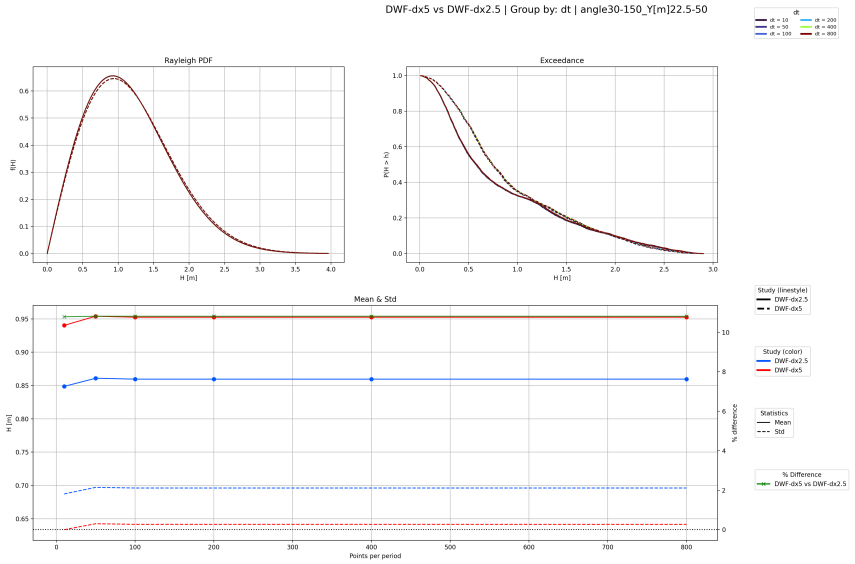


Figure 4.14: Direct comparison between $\Delta x = 5$ m and $\Delta x = 2.5$ m across all conditions. The distributions remain nearly identical, while a systematic $\sim 10\%$ deviation is observed in the mean wave height.

The largest deviations are observed for oblique wave angles (approximately $90°$–$120°$), where increased wave interaction and stronger spatial gradients occur. In these cases, the coarser grid tends to smooth the solution, leading to a slight underrepresentation of peak amplitudes.

Refining the grid from $\Delta x = 5\,\mathrm{m}$ to $2.5\,\mathrm{m}$ yields an improvement in accuracy of approximately 8–15%, depending on the angle, whereas further refinement to $\Delta x = 1\,\mathrm{m}$ provides only marginal gains (typically below 5%).

This behaviour is clearly illustrated in Figure 4.16, where the statistical distributions remain consistent, but the mean wave height exhibits systematic deviations. The overall difference is around 10%, with local peaks reaching up to 15–18% for specific oblique directions.

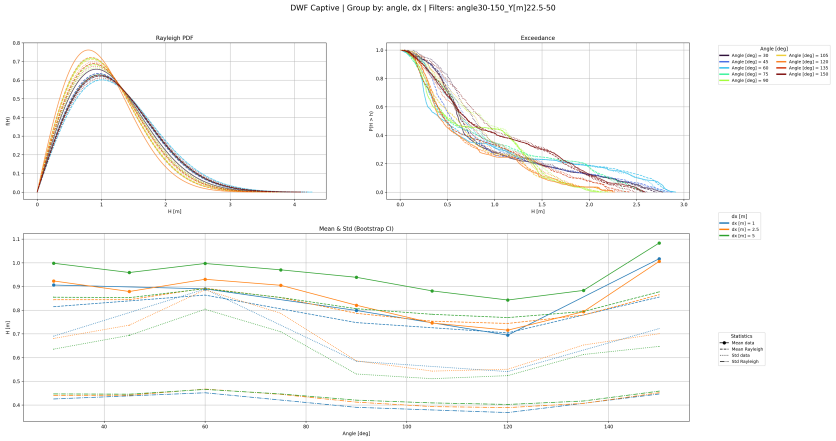


Figure 4.15: Wave height statistics as a function of wave angle for different spatial resolutions.

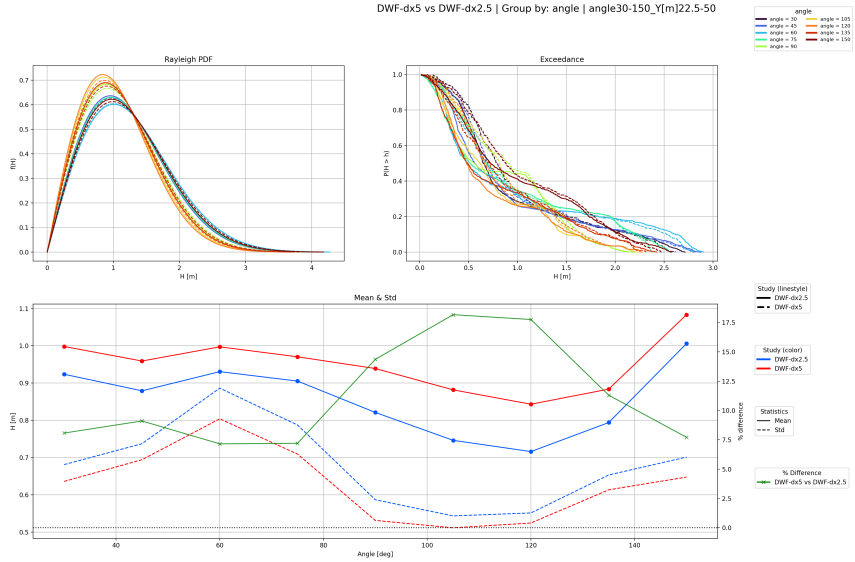


Figure 4.16: Direct comparison between $\Delta x = 5$ m and $\Delta x = 2.5$ m as a function of wave angle. Deviations in mean wave height remain moderate but increase for oblique directions.

Dependence on Wave Frequency A stronger dependence on spatial resolution is observed when considering wave frequency, as illustrated in Figure 4.17. High-frequency waves, characterized by shorter wavelengths, require finer spatial discretization to be accurately resolved.

At low frequencies ($\omega \lesssim 0.5$ rad/s), all grid resolutions yield nearly identical results, confirming that long waves are well resolved even on coarse grids. However, discrepancies increase progressively with frequency.

As shown in Figure 4.18, deviations remain below 5% at low frequencies, increase to 5–10% in the intermediate range ($\omega \approx 0.8$–1.2 rad/s), and exceed 20%–30% at higher frequencies. These differences are associated with the decreasing number of grid points per wavelength, which leads to insufficient spatial resolution for short waves in the $\Delta x = 5$ m case.

Dependence on Spatial Location The influence of spatial resolution also depends on probe location, as shown in Figure 4.19. The largest discrepancies are observed in the central region, where differences reach approximately 0.2 m in wave height (up to 50% relative error). In contrast, the $\Delta x = 2.5$ m grid provides

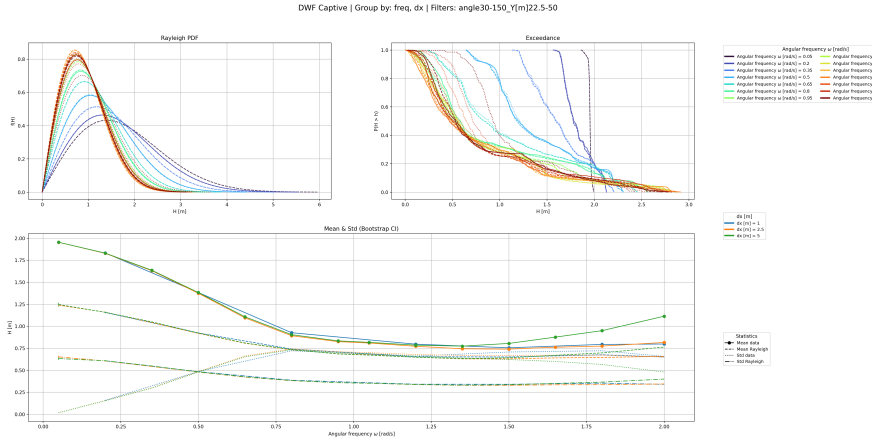


Figure 4.17: Effect of spatial discretization across different wave frequencies. Higher-frequency components exhibit increased sensitivity to grid resolution.

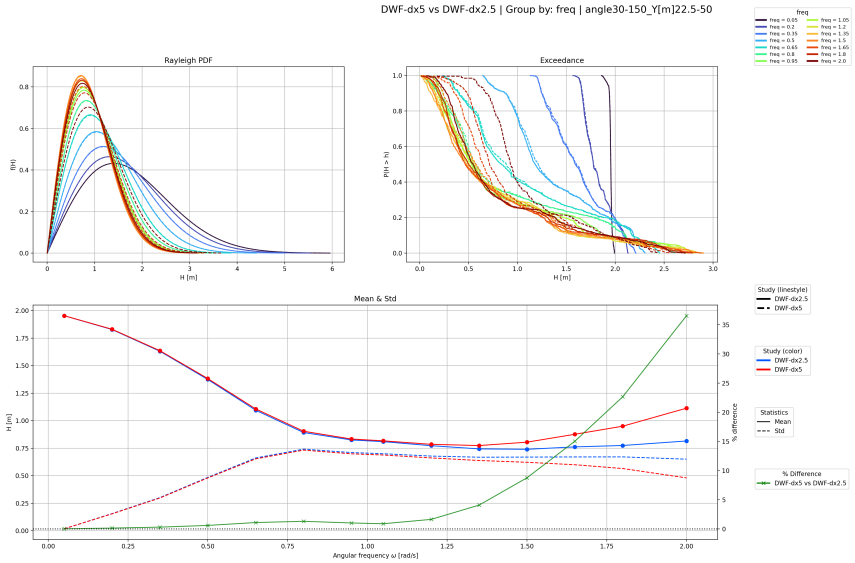


Figure 4.18: Direct comparison between $\Delta x = 5$ m and $\Delta x = 2.5$ m as a function of wave frequency. Errors increase significantly for high-frequency waves.

converged results across most of the domain.

This behaviour is further highlighted in Figure 4.20, which shows separate comparisons for different wave direction ranges. The central region and the half-plane opposing wave incidence exhibit deteriorated agreement, consistent with the presence of shorter wavelengths and stronger interference patterns. These effects are attributed to the shielding of the wave field by the vessel, which locally increases spatial gradients and reduces the effective wavelength.

In the transverse direction (Figure 4.21), the influence of spatial resolution is significantly weaker. Both $\Delta x = 2.5\,\mathrm{m}$ and $\Delta x = 1\,\mathrm{m}$ yield nearly identical results. Close to the vessel, differences of up to $0.2\,\mathrm{m}$ ($\sim 25\%$) are observed, but these diminish to below $0.1\,\mathrm{m}$ (~ 10–12%) further away.

Summary and Recommendations The spatial sensitivity analysis demonstrates that the influence of grid resolution on the simulated wave field is moderate at the global level but becomes significant under specific conditions.

Globally, refining the grid from $\Delta x = 5\,\mathrm{m}$ to $2.5\,\mathrm{m}$ reduces errors to within approximately 10%, while further

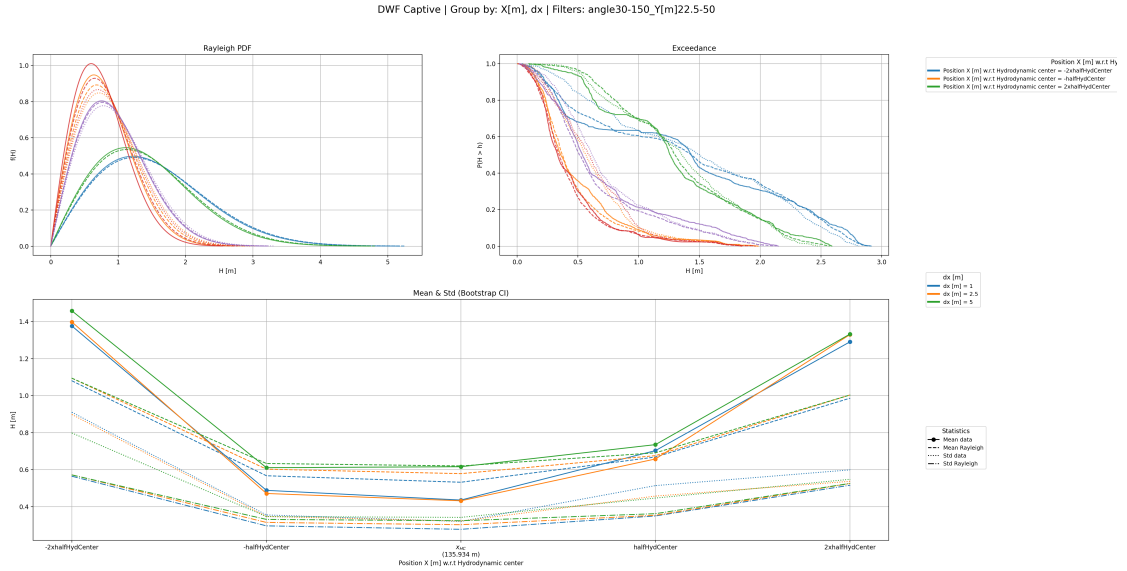


Figure 4.19: Wave height statistics along the longitudinal direction (x).

refinement to $\Delta x = 1\,\text{m}$ yields only marginal improvements ($< 5\%$). The dominant source of sensitivity is associated with high-frequency waves and regions with strong spatial gradients.

From a physical perspective, the required spatial resolution is governed by the shortest wavelength present in the system. For the considered frequency range ($\omega = 0.05$–$2\ \text{rad/s}$), a grid size of $\Delta x = 2.5\,\text{m}$ provides approximately 4–6 grid points per wavelength for the shortest waves, which is sufficient to achieve near-converged results.

The coarsest grid ($\Delta x = 5\,\text{m}$), corresponding to only 2–3 points per wavelength for high-frequency waves, is adequate for low-frequency analyses but introduces significant errors for shorter wavelengths and in high-gradient regions.

The finest grid ($\Delta x = 1\,\text{m}$), providing more than 10 points per wavelength, ensures high accuracy but offers limited additional benefit compared to the intermediate grid.

Overall, a spatial resolution of $\Delta x = 2.5\ \text{m}$ represents an optimal compromise between computational efficiency and accuracy, capturing the essential physics of the wave field while maintaining manageable computational cost. The results also confirm that, once temporal resolution is sufficiently refined, spatial discretization becomes the dominant factor controlling numerical accuracy.

4.2.4 Influence of Vertical Grid Spacing

An additional parameter of interest is the vertical spacing between the two grids used in the DIFFRAC computations. To assess its impact on the resulting wave field, a comparative study was performed using two configurations with vertical separations of $\Delta z = 0.1\ \text{m}$ and $\Delta z = 1\ \text{m}$.

In all cases, the wave probes were positioned at $z = -10^{-6}\ \text{m}$. This slight negative offset ensures that the probes remain within the fluid domain and consistently capture the wave field without being affected by potential numerical issues at the exact free-surface level.

Based on the results of the previous sensitivity analyses, the comparison was limited to a representative case, as the influence of other parameters (e.g., time step and spatial resolution) has already been shown to be well controlled.

A comparison of the resulting wave amplitude distributions for both configurations is shown in Figure 4.22. The left panel corresponds to the case with $\Delta z = 1\ \text{m}$ (for an incident wave angle of $30° - 150°$), while

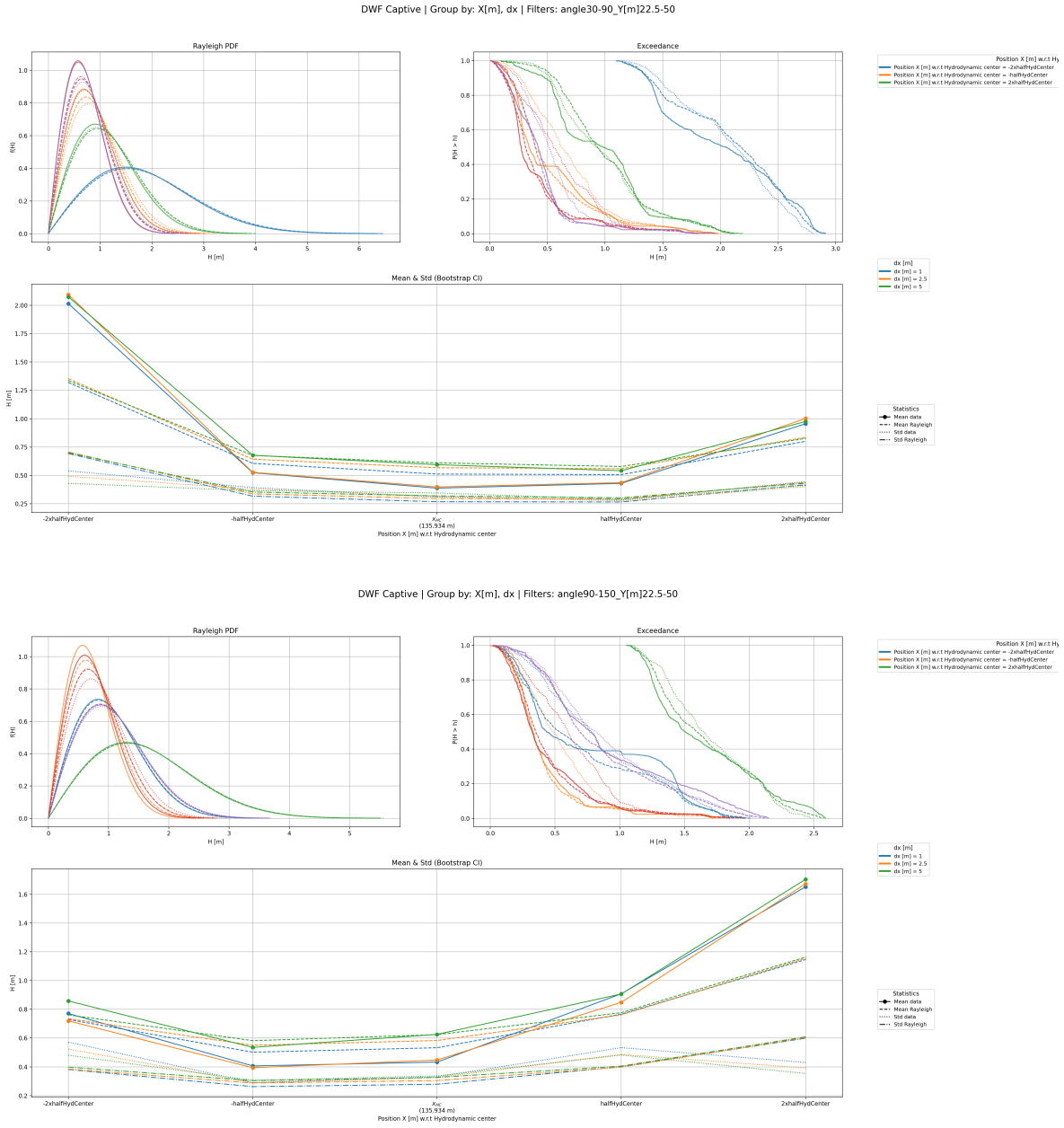


Figure 4.20: Comparison of wave height along the x-direction for different angle ranges.

the right panel shows the results for $\Delta z = 0.1$ m. Both cases were computed using a spatial resolution of $\Delta x = 2.5$ m and identical temporal discretization, with the results averaged over the same number of points per wave period.

The comparison indicates that the influence of the vertical grid spacing on the solution is negligible. The probability distributions and overall wave characteristics are nearly indistinguishable between the two configurations. The maximum observed difference in wave height is approximately 0.1 m, corresponding to a difference of about 0.05 m in crest and trough amplitudes.

Considering that the characteristic wave height is on the order of 1 m, this represents a relative variation of at most 10%, even when the grid spacing is reduced by a factor of 10. This result suggests that the vertical resolution used in the DIFFRAC grid has a limited impact on the accuracy of the simulated wave field for the present configuration.

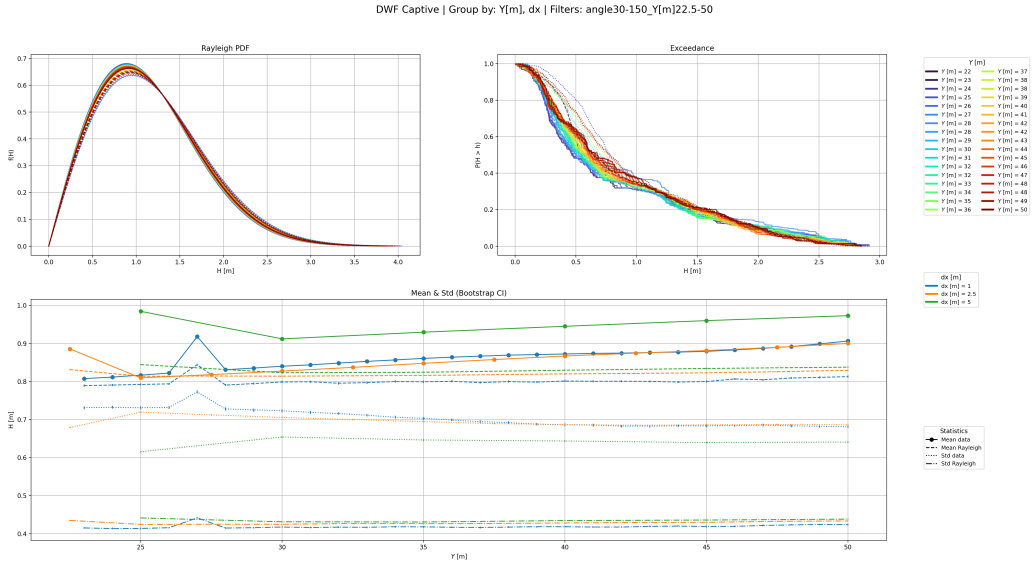
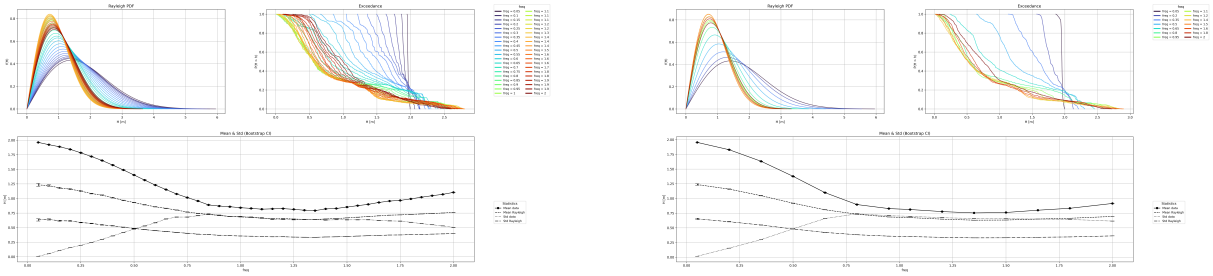


Figure 4.21: Wave height statistics along the transverse direction (y).



(a) $\Delta z = 1\,\mathrm{m}$ *(b)* $\Delta z = 0.1\,\mathrm{m}$

Figure 4.22: Comparison of wave height distributions obtained with two vertical grid spacings in DIFFRAC computations. The left panel shows the case with $\Delta z = 1\,\mathrm{m}$*, while the right panel corresponds to* $\Delta z = 0.1\,\mathrm{m}$*. Both simulations use identical spatial and temporal discretization parameters (*$\Delta x = 2.5\,\mathrm{m}$*) and probe locations at* $z = -10^{-6}\,\mathrm{m}$*. The results demonstrate that reducing the vertical spacing has a negligible impact on the predicted wave field for long waves but around 10% impact on short waves.*

Overall, the results confirm that a vertical spacing of $\Delta z = 1\ \mathrm{m}$ is sufficient to obtain reliable results, and further refinement does not lead to significant improvements in the statistical properties of the solution, but for short waves a smaller refinement is advisable.

4.3 Angular Interpolation Sensitivity Study

A dedicated sensitivity analysis was conducted to evaluate the accuracy and robustness of the angular interpolation procedure used to reconstruct the wave field for incident directions not explicitly available in the hydrodynamic database. The main objective is to quantify the error introduced by interpolation and to establish practical guidelines for angular discretization.

All simulations were performed using a spatial resolution of $\Delta x = 2.5\ \mathrm{m}$ and a sufficiently refined temporal discretization ($\Delta t \leq T/100$), ensuring that both spatial and temporal errors remain negligible. The analysis focuses on intermediate wave directions (e.g., $45°$, $75°$, $105°$, and $135°$), for which the solution is entirely obtained through interpolation between neighbouring database angles.

The interpolated solutions are systematically compared with the corresponding reference simulations obtained

at discrete angular positions in the database, which serve as a benchmark for assessing interpolation accuracy.

Global Behaviour of Interpolated Solutions The global statistical properties of the interpolated wave field are illustrated in Figure 4.23. The probability density functions (PDFs) and corresponding exceedance curves exhibit smooth and physically consistent behaviour across all considered directions, with no visible artifacts introduced by the interpolation procedure.
A direct comparison with the reference solutions is shown in Figure 4.24. The overlap of the distributions confirms that angular interpolation preserves the overall shape of the wave height statistics. The discrepancy in mean wave height remains small and systematic, with differences typically below 0.05 m, corresponding to approximately 5–6%.

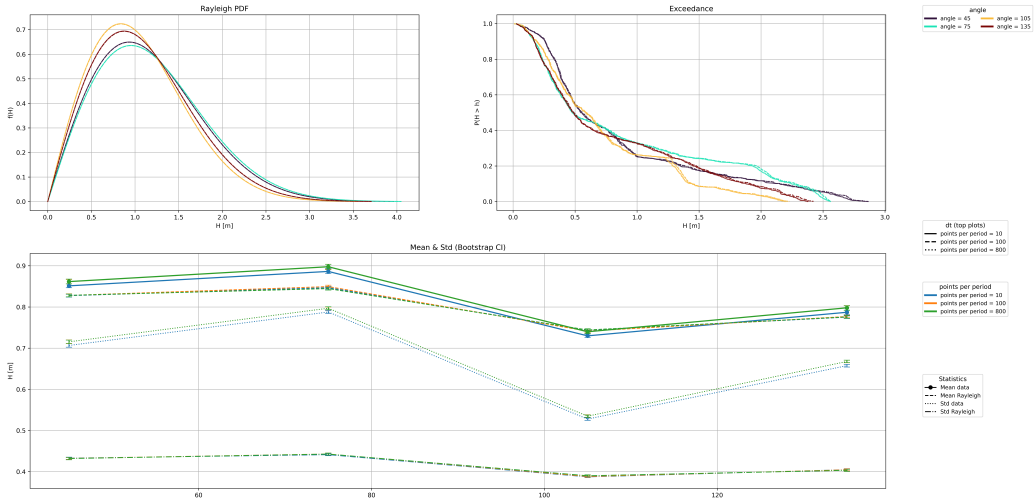


Figure 4.23: Probability density functions of wave height for interpolated wave directions. The distributions remain smooth and free of numerical artifacts.

Dependence on Wave Frequency The effect of interpolation as a function of wave frequency is shown in Figures 4.25 and 4.26. The interpolated solutions reproduce the expected physical trends, with decreasing wave height and steeper exceedance curves as frequency increases.
At low frequencies, interpolation errors are negligible. However, at higher frequencies, where wave propagation becomes more sensitive to direction, deviations increase. As shown in Figure 4.26, the difference in mean wave height remains below 10% in most of the range but can locally approach 15% for the highest frequencies. This reflects the increasing importance of directional resolution for short-wavelength components.

Dependence on Spatial Location The spatial dependence of the interpolation error is analysed along both the transverse (y) and longitudinal (x) directions.
In the transverse direction (Figures 4.27 and 4.28), the interpolated results closely follow the reference solutions. The largest deviations occur near the vessel, where the wave field is more complex, reaching approximately 5–7%. However, these differences rapidly decrease with distance, indicating that the interpolation error is secondary compared to the physical spatial variation of the wave field.

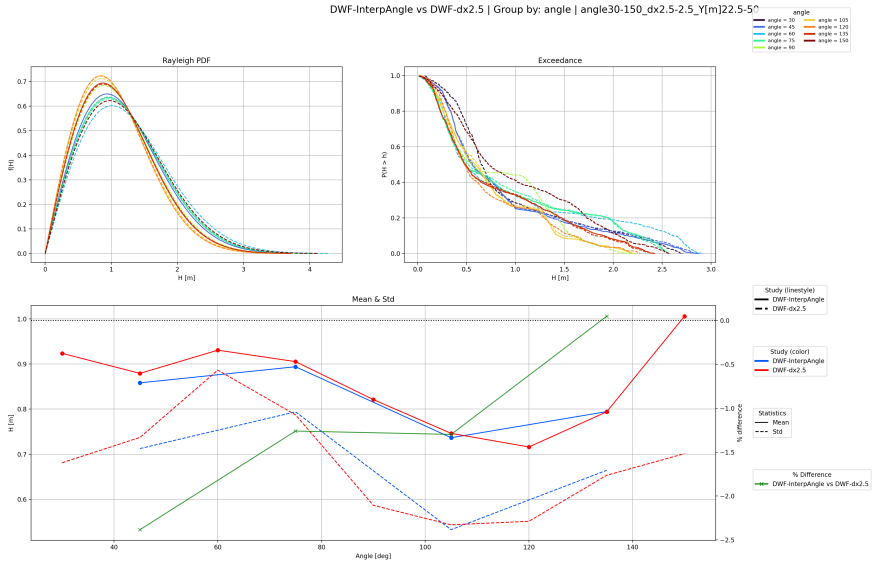


Figure 4.24: Comparison between interpolated and reference solutions as a function of wave direction. Deviations in mean wave height remain within approximately 5–6%.

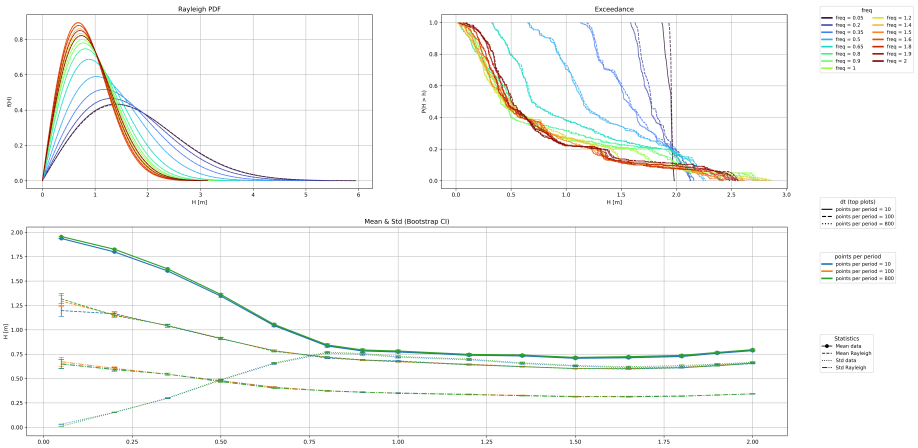


Figure 4.25: Wave height statistics for interpolated directions across frequencies. The physical trends are preserved.

The longitudinal direction exhibits a more complex behaviour, as shown in Figures 4.29, 4.30, and 4.31. The largest discrepancies are observed in the central region, where wave interference and shielding effects generate shorter effective wavelengths.

At the hydrodynamic center, interpolation leads to deviations that can locally exceed 20% and reach up to 30%–40% in extreme cases. This is not due to interpolation instability but rather to the strong sensitivity of this region to directional changes. Small variations in angle produce significant changes in wave interaction patterns, which are difficult to reconstruct linearly.

Away from the center, the differences reduce substantially and remain within approximately 10–15%. The comparison of different angle ranges confirms that the largest discrepancies occur in the half-plane opposite to the incoming waves, where shielding effects dominate and spatial gradients are strongest.

Summary and Recommendations The angular interpolation procedure provides a stable and physically consistent reconstruction of the wave field for intermediate directions. At the global level, interpolation errors

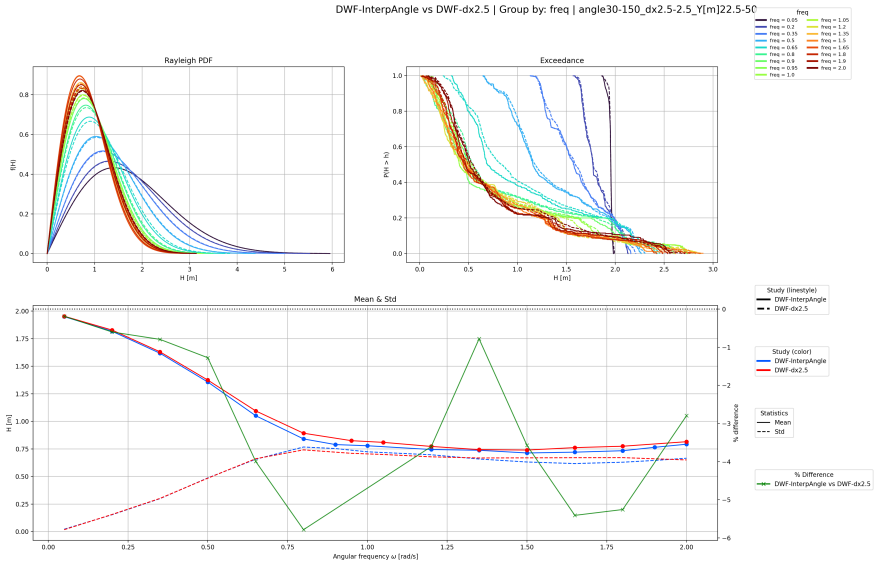


Figure 4.26: Comparison between interpolated and reference solutions as a function of frequency. Errors increase with frequency, reaching up to approximately 10–15% at the highest values.

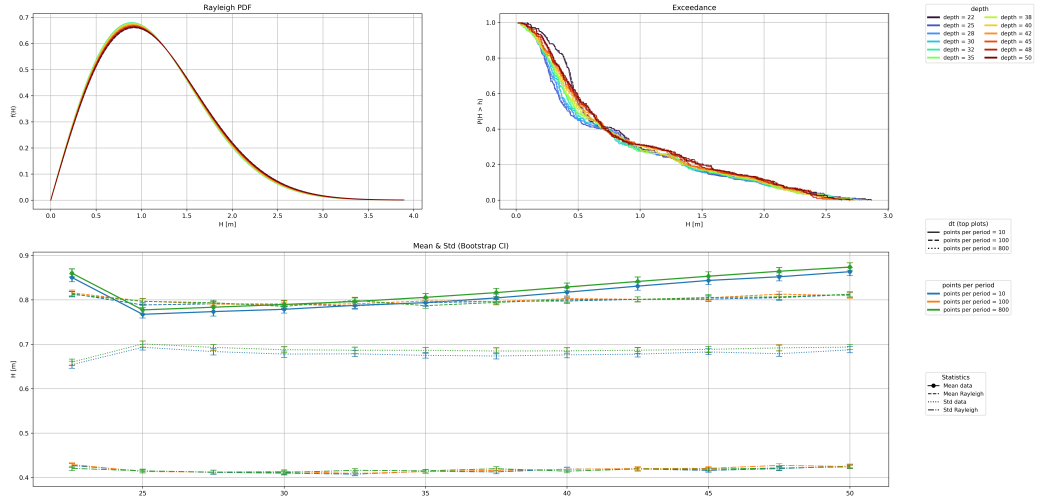


Figure 4.27: Wave height statistics for interpolated angles along the transverse direction.

remain small, typically within 5–6%, and do not affect the overall statistical behaviour of the solution.

The sensitivity of the interpolation is primarily governed by two factors: wave frequency and spatial location. High-frequency waves and regions with strong spatial gradients, such as the central wake region, exhibit increased sensitivity to directional variations, leading to larger discrepancies.

Despite these local deviations, the interpolation method remains robust and reliable, with errors significantly smaller than those associated with coarse spatial discretization. The results demonstrate that an angular spacing of 30° is sufficient to accurately reconstruct intermediate wave directions for most engineering applications.

However, for cases involving strong directional sensitivity or high-frequency wave components, a finer angular discretization may be required to accurately capture local wave phenomena. Overall, the interpolation approach provides an efficient and accurate solution, enabling substantial reductions in computational cost

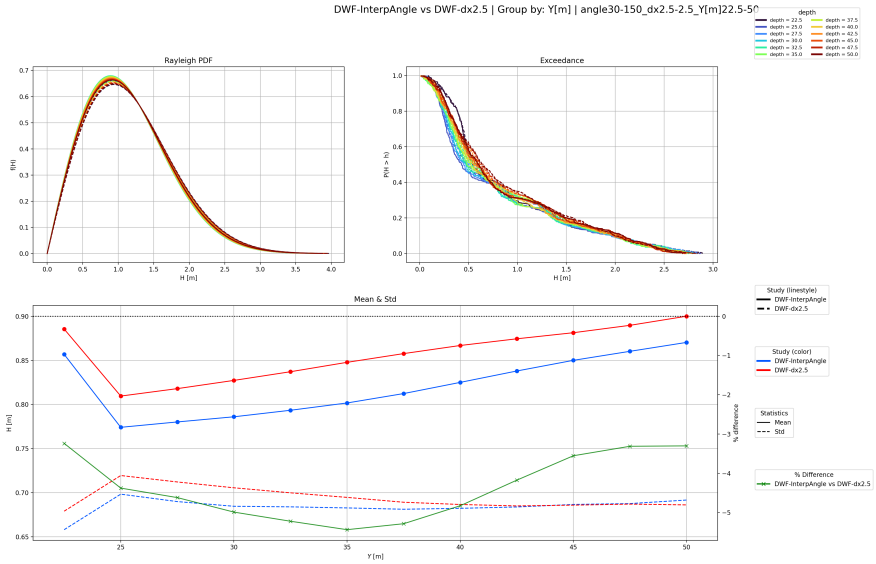


Figure 4.28: Comparison between interpolated and reference solutions along the transverse direction.

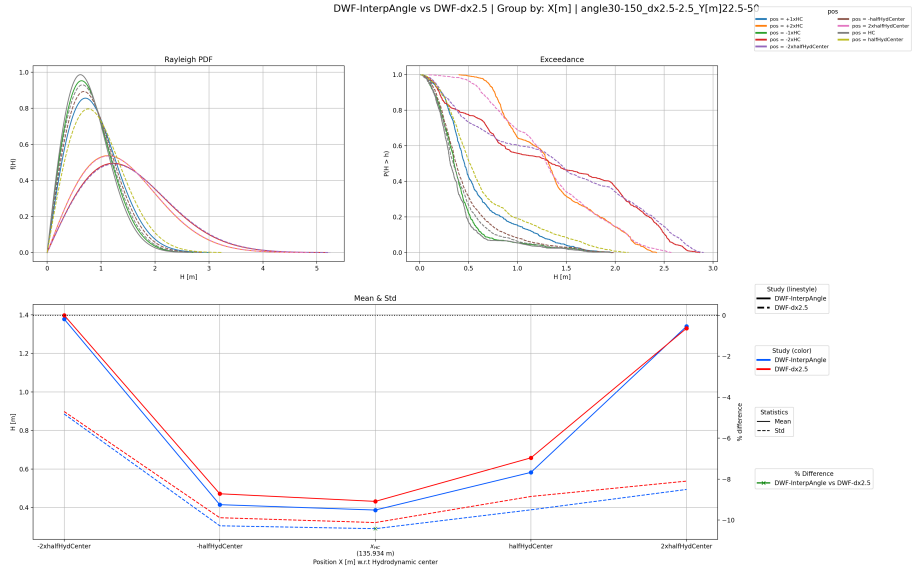


Figure 4.29: Comparison between interpolated and reference solutions along the longitudinal direction for the full angular range.

without compromising the integrity of the results.

4.4 Interpolation Spatial Step Study

In this section, the influence of spatial interpolation on the evaluation of the wave field is analysed. The wave probe locations are shifted by $1\,\mathrm{m}$ relative to the computational grid, requiring the Discrete Wave Field (DWF) model to interpolate the free-surface elevation at off-grid positions. All simulations are performed using a spatial resolution of $\Delta x = 2.5\,\mathrm{m}$, which was previously identified as providing converged results.

To quantify the interpolation error, the results are compared with a reference configuration in which probe locations coincide exactly with grid nodes, thus eliminating any interpolation. The comparison is performed over the full range of wave incidence angles, frequencies, and spatial locations considered in the study.

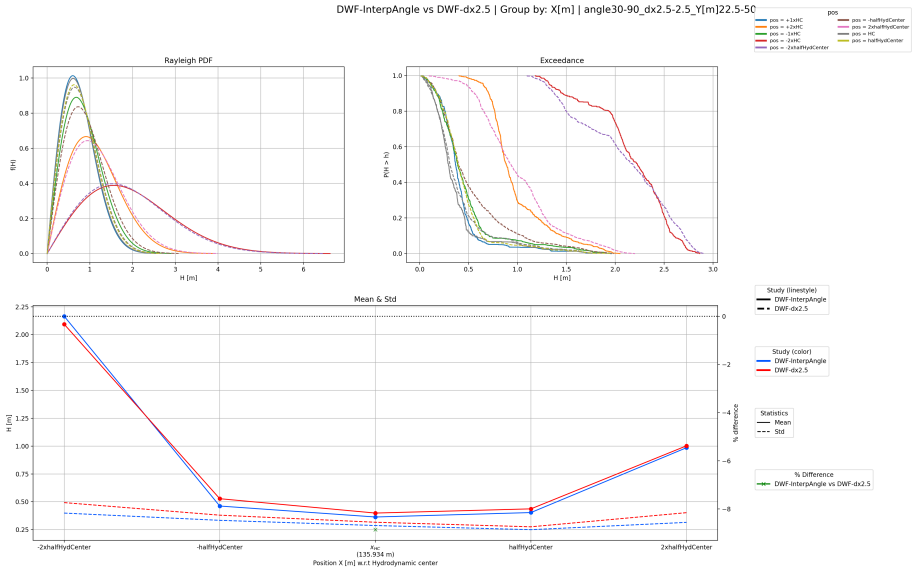


Figure 4.30: Longitudinal comparison for wave angles between $30°$ and $90°$.

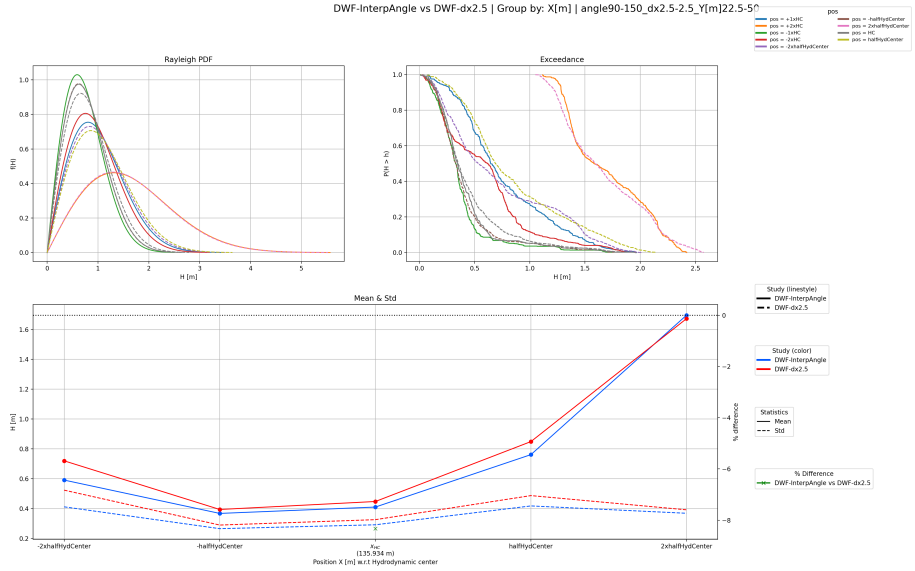


Figure 4.31: Longitudinal comparison for wave angles between $90°$ and $150°$.

Global Behaviour The global impact of spatial interpolation is illustrated in Figure 4.32. The probability density functions (PDFs) and exceedance curves for the interpolated and non-interpolated solutions exhibit a very close agreement, with nearly overlapping distributions across all wave directions.
The mean wave height shows only a small systematic deviation, typically within 3–5%, indicating that the interpolation procedure does not significantly alter the global statistical properties of the wave field. This confirms that spatial interpolation introduces only minor, well-controlled errors at the global level.

Dependence on Transverse Position The behaviour of the interpolation along the transverse (y) direction is shown in 4.33. The interpolated results closely match the reference solution over the entire domain.
Differences in wave height remain below $0.05\,\text{m}$, corresponding to approximately 2–5%. Slightly larger deviations are observed closer to the vessel, where the wave field exhibits stronger gradients. However, these differences remain small compared to the natural spatial variability of the wave field, confirming that

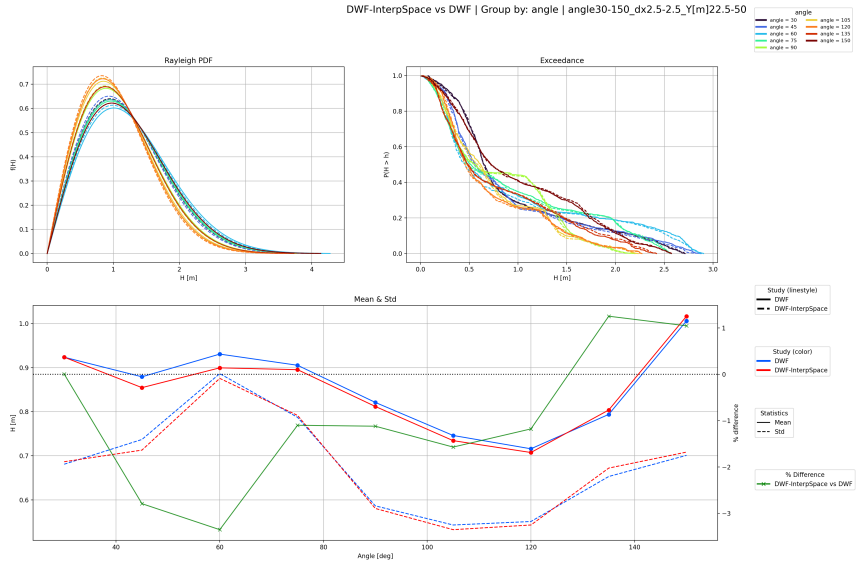


Figure 4.32: Global comparison between interpolated and non-interpolated solutions as a function of wave direction. The distributions remain nearly identical, with mean deviations typically below 5%.

the interpolation error is secondary to the underlying physical variation.

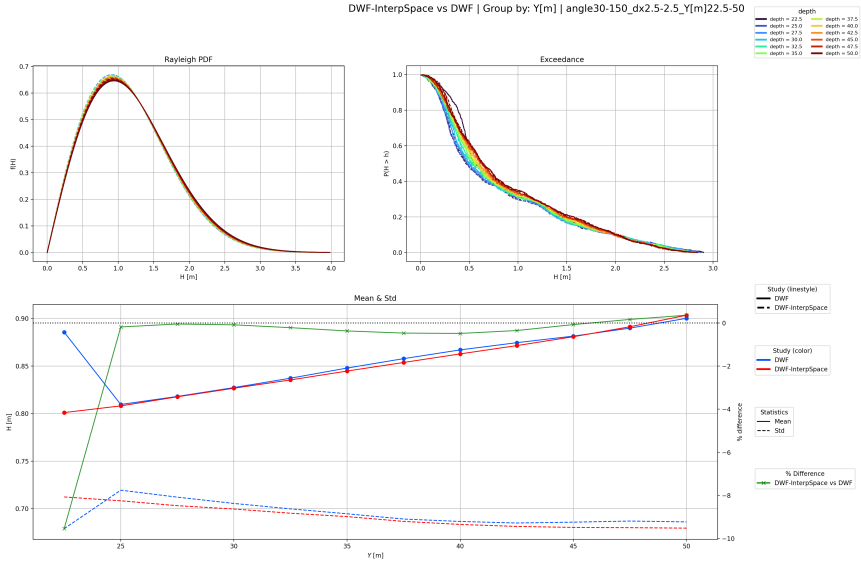
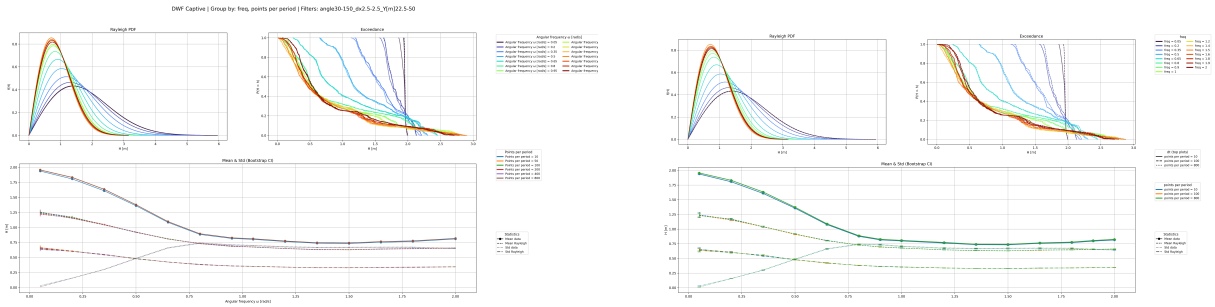


Figure 4.33: Direct comparison along the transverse direction. Deviations remain small and nearly uniform across the domain.

Dependence on Wave Frequency The sensitivity of spatial interpolation to wave frequency is presented in Figures 4.34 and 4.35. At low frequencies, the interpolated and reference solutions are virtually indistinguishable.

As the frequency increases, small discrepancies emerge, particularly in the tail of the distributions. These differences remain limited, generally below 5%, although local deviations may increase slightly for the highest frequencies. This behaviour is consistent with the reduced number of effective grid points per wavelength at higher frequencies, which increases sensitivity to interpolation.



(a) Reference solution *(b) Interpolated solution*

Figure 4.34: Frequency-domain comparison between interpolated and reference solutions.

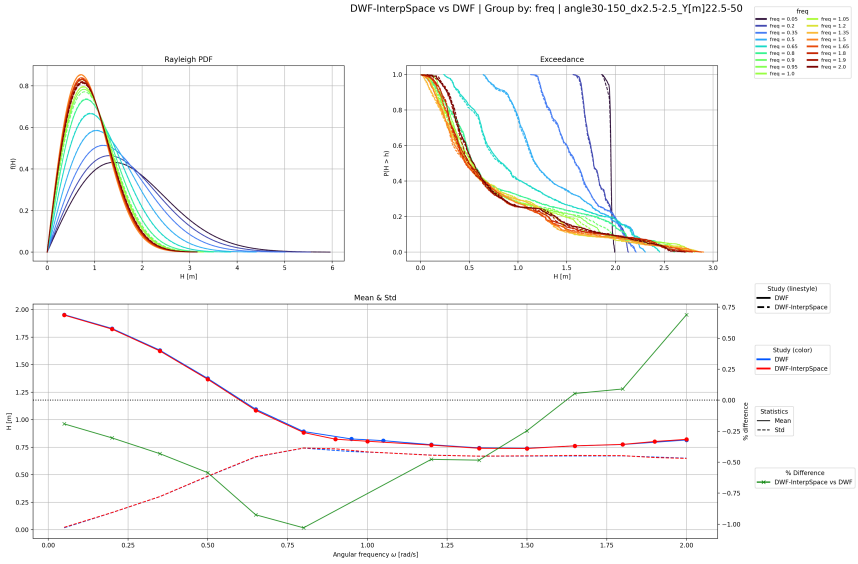


Figure 4.35: Direct comparison as a function of wave frequency. Differences increase slightly for high-frequency components.

Dependence on Longitudinal Position The effect of interpolation along the longitudinal (x) direction is shown in Figures 4.36, 4.37, and 4.38. Compared to the transverse direction, the sensitivity to interpolation is more pronounced due to the presence of stronger wave gradients and interference effects.
In particular, the largest deviations occur near the hydrodynamic center, where the wave field exhibits strong constructive and destructive interference. In this region, relative differences can locally increase, reaching values on the order of 10–20%, although the absolute differences remain limited (typically below 0.05–0.1 m). Away from the central region, the discrepancies decrease significantly and remain within approximately 5–10%. The decomposition into different angle ranges shows that the largest differences occur in the half-plane opposing the incident wave direction, where shielding and wave interaction effects are strongest.
These results indicate that the interpolation error is primarily controlled by the local complexity of the wave field rather than by the interpolation procedure itself.

Summary and Recommendations The analysis demonstrates that spatial interpolation introduces only minor and well-controlled errors in the evaluation of the wave field at off-grid locations. At the global level, deviations remain small, typically below 5% in wave height, and do not affect the overall statistical behaviour. The interpolation error exhibits a moderate dependence on frequency and spatial location, with slightly increased sensitivity for high-frequency waves and in regions of strong wave interaction, such as the central

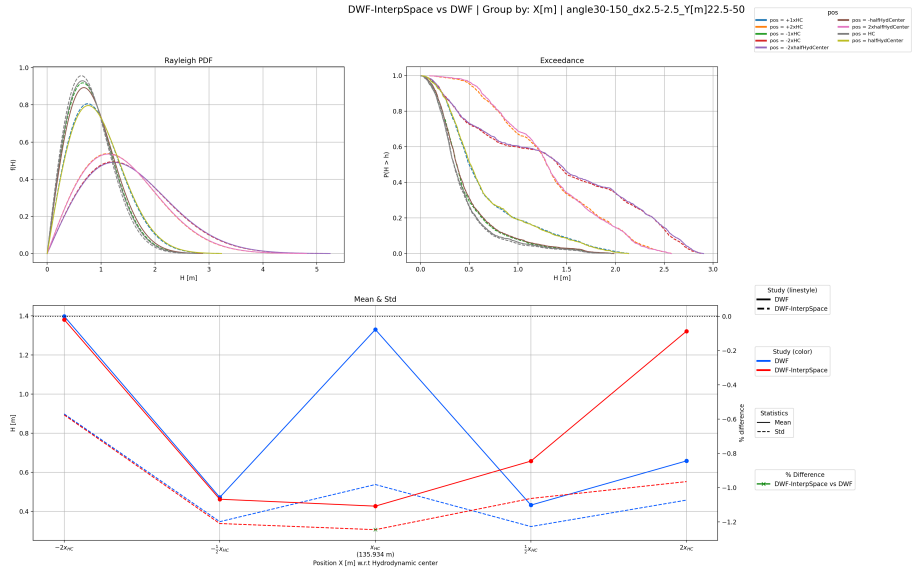


Figure 4.36: Comparison between interpolated and reference solutions along the longitudinal direction for the full angular range.

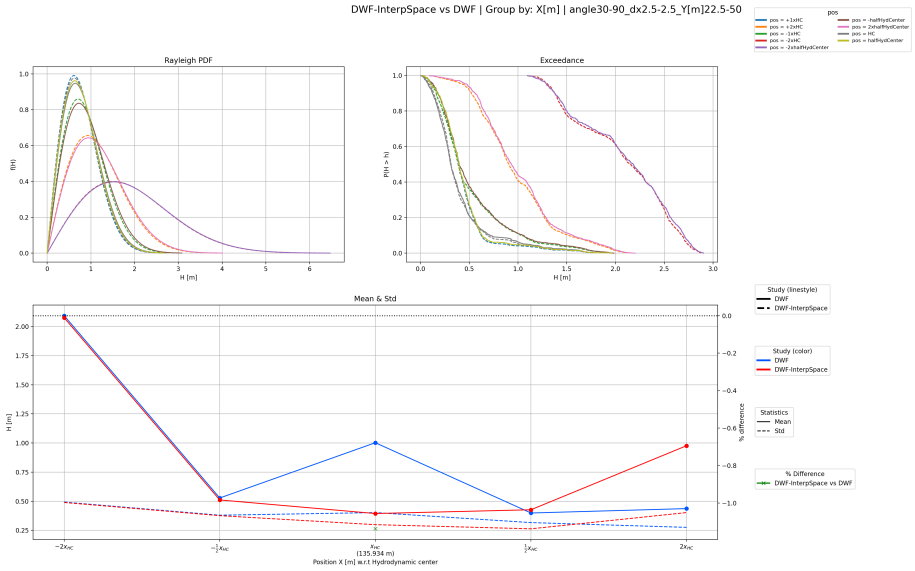


Figure 4.37: Longitudinal comparison for wave angles between $30°$ and $90°$.

wake region. However, even in these cases, the interpolation error remains significantly smaller than the variations induced by coarse spatial discretization.

Overall, the results confirm that spatial interpolation at $\Delta x = 2.5\ \mathrm{m}$ provides a reliable and accurate method for evaluating wave probes at arbitrary locations. This enables flexible probe placement without compromising solution quality, making the approach suitable for practical engineering applications.

4.5 Interpolation Frequency Study

In this analysis, we look into the frequency interpolation node of DWF.

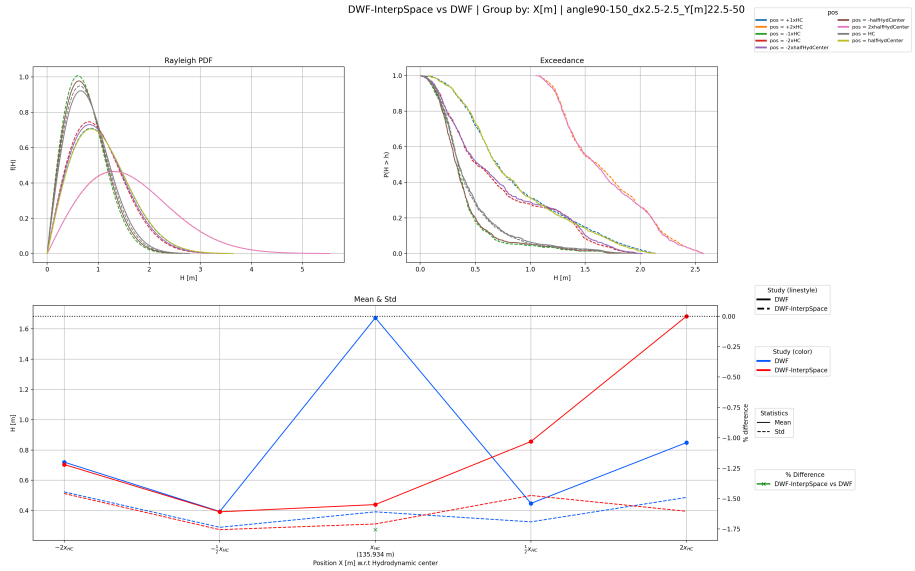


Figure 4.38: Longitudinal comparison for wave angles between $90°$ and $150°$.

4.6 Comparison Against VBM-XMF

A comprehensive comparison is performed between the Discrete Wave Field (DWF) model and the VBM-XMF approach to evaluate the consistency in predicting diffraction wave fields. The analysis covers a wide range of wave incidence angles ($30°$–$150°$), frequencies ($\omega \geq 0.5$ rad/s), and spatial probe locations, allowing both global trends and local discrepancies to be assessed.

Global Behaviour The comparison as a function of wave direction is presented in Figure 4.39. The probability density functions (PDFs) and exceedance curves obtained from both models exhibit a high degree of similarity. The overall shape of the distributions and the variation with angle are consistently reproduced, confirming that both approaches capture the same physical wave processes.
Despite this agreement in statistical structure, a clear systematic deviation is observed in the predicted wave heights. The VBM-XMF model consistently produces lower mean values compared to DWF. As shown in the lower panel of Figure 4.39, the relative difference generally lies within 10–20%, with slightly larger deviations for intermediate angles where wave interaction effects are more pronounced.

Dependence on Transverse Position The comparison along the transverse (y) direction is shown in Figure 4.40. Both models display similar spatial trends, with smooth variations of the wave height as the distance from the vessel increases.
The absolute differences between the two solutions typically range between 0.1 and 0.2 m. When normalised with respect to the local wave height, this corresponds to relative differences of approximately 10–20%. The discrepancy is relatively uniform across the transverse direction, indicating that the deviation is linked to the modelling approach rather than local spatial effects.

Dependence on Wave Frequency The frequency-dependent comparison is presented in Figure 4.41. At lower frequencies, both models show good agreement, with only minor differences in wave height and statistical distribution.
However, as the frequency increases, the discrepancy becomes more pronounced. For $\omega \gtrsim 0.75$ rad/s, the VBM-XMF model underestimates the wave height more significantly, with deviations reaching 15–25% and

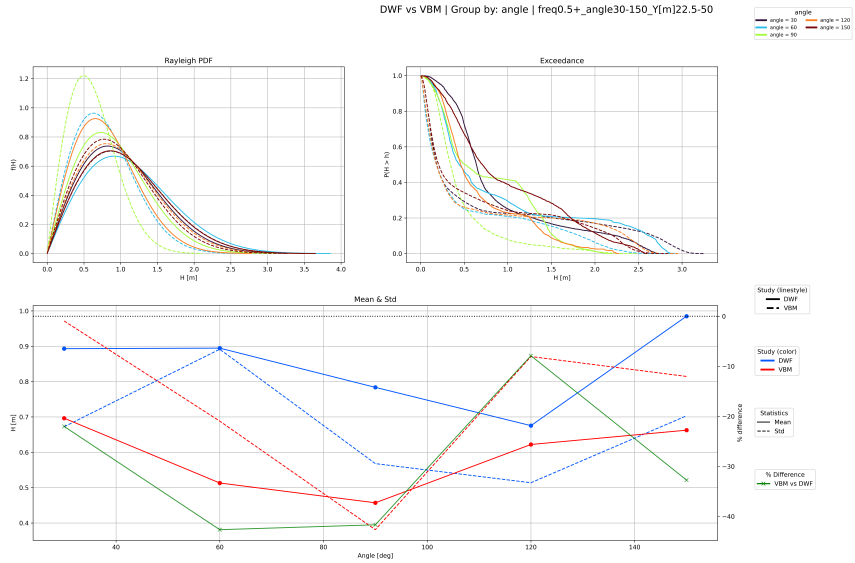


Figure 4.39: Comparison between DWF and VBM-XMF as a function of wave incidence angle. The statistical distributions are consistent, while VBM-XMF systematically underestimates the mean wave height by approximately 10–20%.

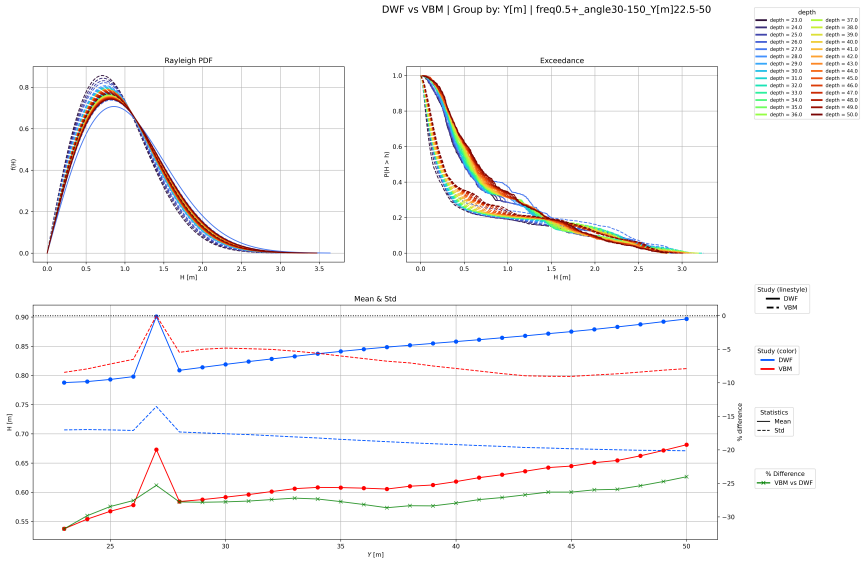


Figure 4.40: Comparison between DWF and VBM-XMF along the transverse direction. Both models follow similar spatial trends, with VBM-XMF predicting consistently lower values.

locally exceeding this range at the highest frequencies.

This behaviour highlights the increased sensitivity of short-wavelength waves to the underlying modelling assumptions. The DWF method captures higher-frequency components more accurately, while the VBM-XMF approach produces a smoother response, leading to lower predicted amplitudes.

Dependence on Longitudinal Position A more detailed comparison is performed along the longitudinal (x) direction, where stronger spatial gradients and wave interactions occur. The results are presented in Figures 5.66, 4.43, and 4.44.

Along this direction, the discrepancy between the two models becomes more pronounced and spatially

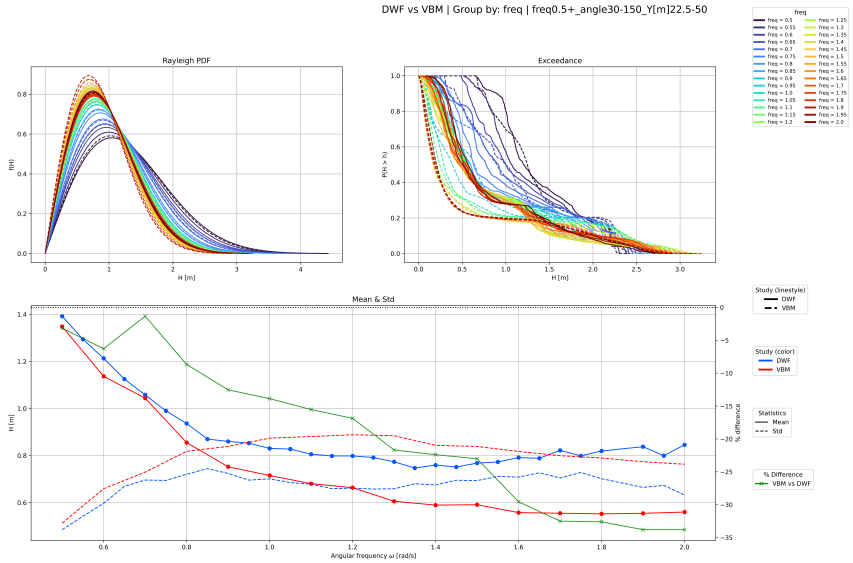


Figure 4.41: Comparison between DWF and VBM-XMF as a function of frequency. Differences increase for higher frequencies due to increased sensitivity of short waves.

dependent. In particular, in regions close to the hydrodynamic centre, where wave interference and shielding effects dominate, the relative difference can locally reach values exceeding 30%–50%. This is especially evident for certain angle ranges, as shown in Figures 4.43 and 4.44, where the differences strongly depend on the direction of wave incidence.

In contrast, away from the central region, the differences decrease significantly, typically remaining within 10–20%. The largest discrepancies are observed in the half-plane opposing the incident wave direction, where complex diffraction patterns and shorter effective wavelengths make the solution highly sensitive to modelling assumptions.

This behaviour indicates that the divergence between DWF and VBM-XMF is primarily driven by the ability of each model to resolve local wave interactions rather than by global wave characteristics.

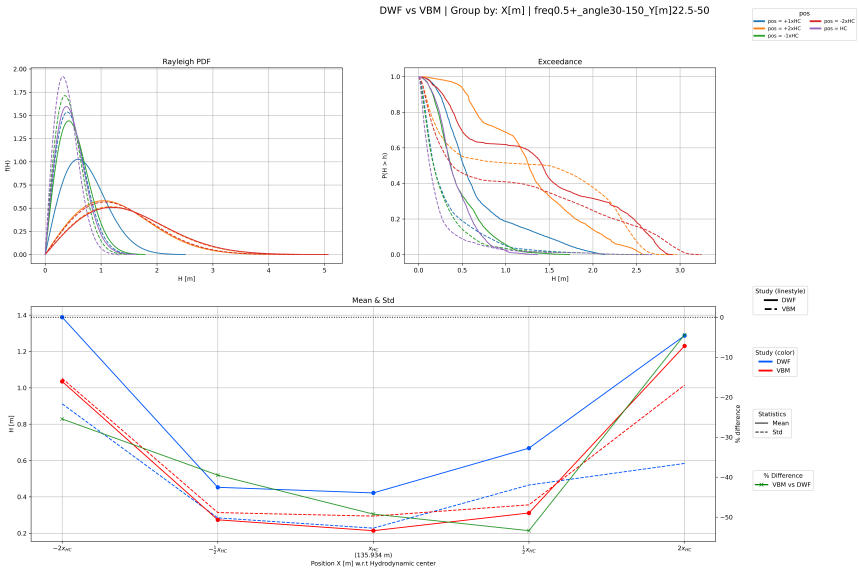


Figure 4.42: Comparison along the longitudinal direction for the full angular range. The largest discrepancies occur near the hydrodynamic centre.

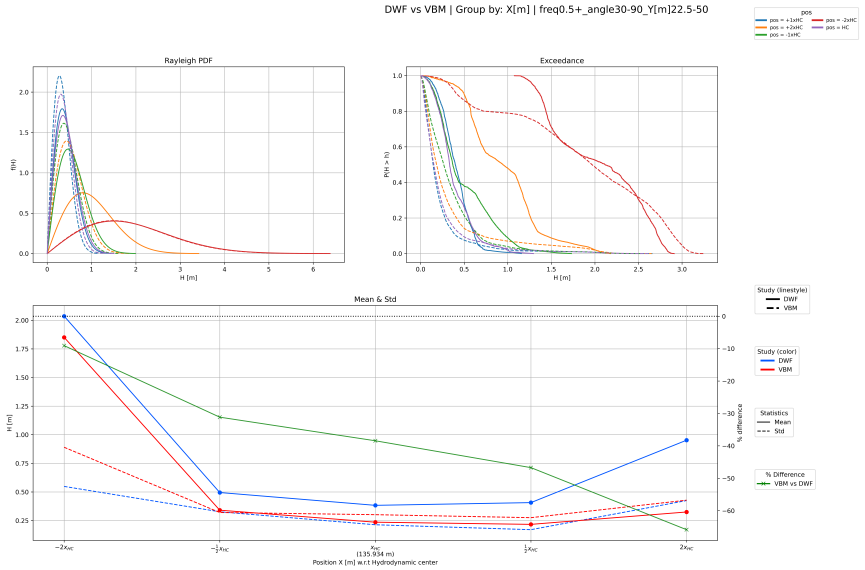


Figure 4.43: Longitudinal comparison for wave angles between $30°$ and $90°$.

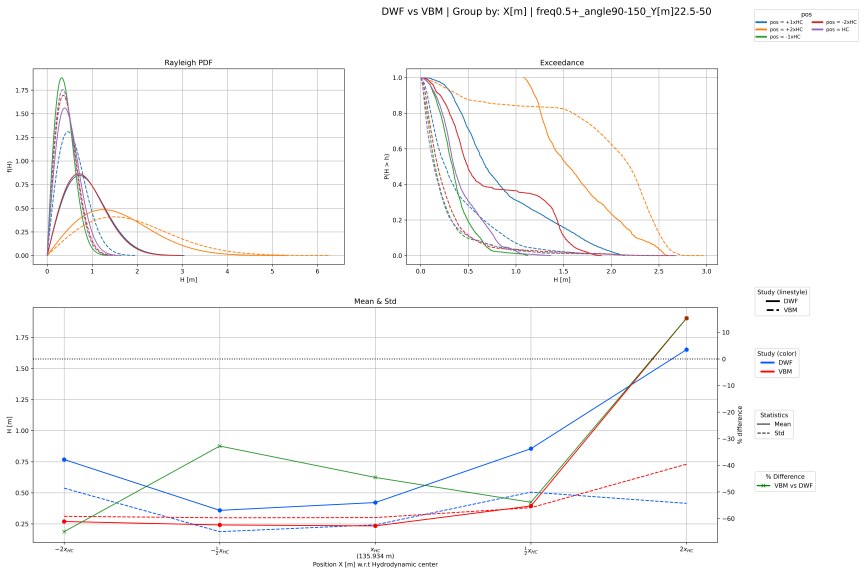


Figure 4.44: Longitudinal comparison for wave angles between $90°$ and $150°$.

Discussion Across all comparisons, both DWF and VBM-XMF reproduce consistent qualitative trends in wave propagation, spatial variation, and directional dependence. This confirms that both models capture the governing physical mechanisms of wave diffraction.

The main difference lies in the amplitude prediction. The VBM-XMF model systematically underestimates the wave height compared to DWF. This discrepancy increases in regions with strong wave interactions and for higher-frequency components.

From a physical perspective:

- The DWF model provides a higher spatial resolution of the wave field, allowing sharper gradients and stronger interference patterns to be resolved,
- The VBM-XMF model introduces a smoother representation of the wave field, which reduces peak amplitudes and dampens local extrema.

Conclusions Overall, the comparison shows that DWF and VBM-XMF provide consistent predictions in terms of global trends and spatial distributions. However, the VBM-XMF model systematically predicts lower wave heights, with differences generally in the range of 10–20%, and locally reaching up to 30–50% in regions with strong wave interactions.
These results indicate that while VBM-XMF is suitable for capturing the overall behaviour of diffraction waves, the DWF approach offers improved accuracy in resolving local wave amplitudes, particularly for high-frequency components and near-field regions.
Consequently, the choice between the two models should depend on the level of accuracy required and the importance of capturing local wave effects in the application of interest.

4.7 Conclusion and Recommendations

This chapter has presented a comprehensive investigation of diffraction wave modelling using the Discrete Wave Field (DWF) approach, including sensitivity analyses, interpolation assessments, and a detailed comparison against the VBM-XMF model. The objective was to quantify numerical accuracy, identify dominant error sources, and provide practical guidelines for reliable simulations.

Numerical Convergence and Dominant Error Sources The sensitivity studies clearly demonstrate that the numerical solution is only weakly dependent on temporal discretization once a moderate resolution is achieved. A time step corresponding to $\Delta t \leq T/100$ (approximately 100 points per wave period) is sufficient to ensure convergence across all tested conditions. Further refinement yields negligible improvements, typically below 5%, indicating that temporal errors are not the limiting factor in the present simulations.
In contrast, spatial discretization plays a significantly more important role. The analysis shows that the required resolution is primarily governed by the number of grid points per wavelength. A spacing of $\Delta x = 2.5\ \mathrm{m}$ provides approximately 4–6 points per wavelength for the highest frequencies considered, which is sufficient to achieve near-converged results. Coarser grids ($\Delta x = 5\ \mathrm{m}$) introduce global deviations of approximately 10% and local errors exceeding 20–30% for short waves and regions with strong spatial gradients.
The largest discrepancies are consistently observed in the central and wake regions of the vessel, where wave interference, diffraction, and shielding effects generate shorter effective wavelengths and steeper gradients. These regions are therefore the most demanding in terms of spatial resolution and constitute the primary source of numerical uncertainty.

Role of Interpolation The various interpolation studies (angular, spatial, and indirectly frequency-related) show that interpolation errors remain generally small compared to discretization errors.
Angular interpolation with a 30° spacing introduces deviations of approximately 5–6% globally, increasing up to 10–15% for high-frequency waves or in regions with strong directional sensitivity. Similarly, spatial interpolation (offset probe locations) produces errors typically below 5%, with local increases up to 10–20% in highly nonlinear regions such as the wake.
These results indicate that both angular and spatial interpolation procedures are robust and suitable for practical applications. The interpolation error remains secondary with respect to spatial discretization, provided that the underlying grid resolution is sufficient.
In contrast, frequency interpolation appears to be more sensitive and may lead to larger discrepancies in certain cases, particularly for high-frequency components. Although not fully explored in this study, the observed behaviour suggests that frequency interpolation requires additional investigation and possibly finer discretization in the spectral domain.

Influence of Vertical Resolution The vertical grid spacing in the DIFFRAC database has a limited influence on the predicted wave field. Reducing the spacing from $\Delta z = 1\ \mathrm{m}$ to $0.1\ \mathrm{m}$ leads to negligible differences for long waves and moderate improvements (up to $\sim 10\%$) for short waves. This confirms that a $\Delta z = 1\ \mathrm{m}$ resolution is generally sufficient, except in applications focusing on high-frequency wave components.

Comparison Between DWF and VBM-XMF The comparison between DWF and VBM-XMF shows that both models reproduce consistent global trends in wave propagation, directional dependence, and spatial variation. The statistical distributions (PDFs and exceedance curves) are in good agreement across all analysed cases, confirming that both approaches capture the same underlying physical processes.
However, a systematic difference in amplitude prediction is observed. The VBM-XMF model consistently underestimates the wave height relative to DWF, with typical discrepancies of 10–20% at the global level. This difference increases with frequency and becomes more pronounced in regions with strong wave interactions.
Locally, particularly in the longitudinal direction near the hydrodynamic centre and in the wake region, the discrepancy can reach 30–50%. These large differences are associated with the high sensitivity of these regions to wave interference and diffraction effects, which require a fine spatial resolution to be accurately captured.
From a modelling perspective, this behaviour reflects the different levels of spatial fidelity of the two approaches. The DWF method resolves the wave field with higher accuracy, capturing sharper gradients and stronger local interactions, whereas the VBM-XMF model provides a smoother representation that reduces peak amplitudes.

Overall Assessment The results of this chapter demonstrate that accurate prediction of diffraction waves is primarily controlled by spatial resolution and the ability to resolve short wavelengths and local interference patterns. Temporal discretization and interpolation procedures, when properly configured, contribute only secondary errors.
The DWF method is shown to provide a reliable and robust framework for modelling diffraction waves, offering high accuracy in both global statistics and local wave behaviour. The VBM-XMF approach, while computationally efficient and physically consistent, tends to underpredict wave amplitudes and may be less suitable for applications requiring precise estimation of extreme values or local wave loads.

Practical Guidelines Based on the findings of this study, the following recommendations can be formulated:

- A temporal resolution of $\Delta t \approx T/100$ (50–100 points per period) is sufficient for convergence,
- A spatial resolution of $\Delta x = 2.5\ \mathrm{m}$ provides an optimal balance between accuracy and computational cost,
- At least 4 grid points per wavelength should be ensured for reliable results, particularly for high-frequency waves,
- Angular interpolation with a 30° spacing is adequate for most applications,
- Spatial interpolation of probe locations introduces negligible errors and can be safely used,
- A vertical grid spacing of $\Delta z = 1\ \mathrm{m}$ is sufficient, except for short-wave dominated cases,
- Additional care is required when analysing high-frequency waves and regions near the vessel centre, where sensitivity to spatial resolution and modelling assumptions is highest.

Final Remarks Overall, this work highlights the importance of resolving the correct spatial scales in diffraction wave modelling. While simplified approaches and interpolation techniques can provide efficient solutions, their limitations become evident in regions with strong wave interaction.

The DWF approach, combined with appropriate numerical settings, is therefore recommended for applications where accurate prediction of wave amplitudes and local wave effects is required. Future work should focus on improving frequency interpolation strategies and further investigating the discrepancies observed between different modelling approaches.

5 Diffracted and Radiated Waves

5.1 Introduction

In this section, vessel motion is activated through a linear spring stiffness matrix, enabling the generation of radiated waves in addition to diffraction. The resulting wave field obtained from the Diffraction Wave Field (DWF) node is analysed and compared across different discretization settings.
Compared to the purely diffractive case, the inclusion of radiated waves introduces shorter wavelengths, stronger spatial gradients, and increased wave–structure coupling. These effects significantly enhance the sensitivity of the numerical solution, particularly with respect to spatial resolution.

5.2 Sensitivity Analysis

5.2.1 Temporal Resolution

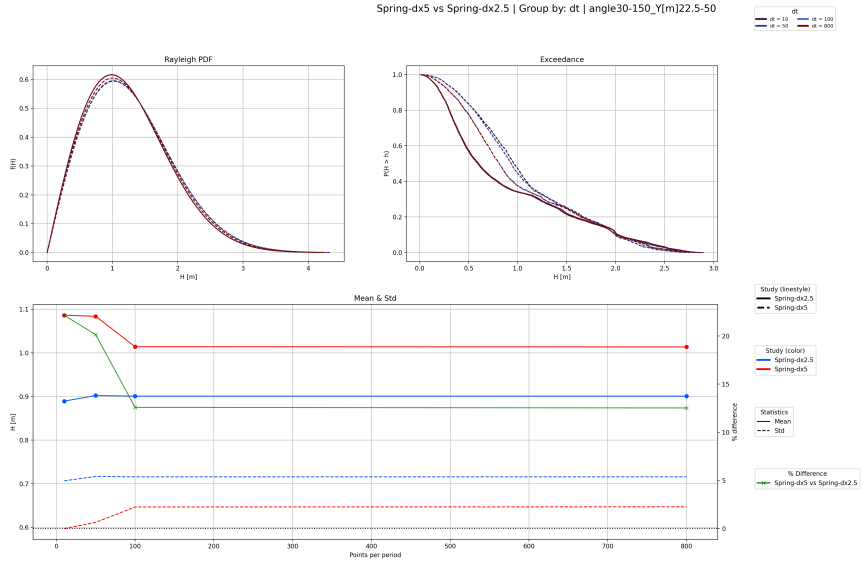


Figure 5.45: Effect of temporal resolution on wave statistics.

Figure 5.45 shows that temporal discretization has a limited impact on the solution. While small deviations are visible at coarse resolution, convergence is rapidly achieved as the number of points per period increases. For $\Delta t \leq T/100$, both the Rayleigh distributions and exceedance curves collapse, and the mean wave height varies by less than 5%.
Recommendation: A temporal resolution of $\Delta t \approx T/100$ is sufficient for accurate simulations. Further refinement yields negligible improvements and is not computationally justified.

5.2.2 Angular Dependence

Figure 5.46 highlights a systematic overestimation of wave height when using the coarse grid ($\Delta x = 5\,\text{m}$). Differences typically range between 10% and 25%, with the largest discrepancies occurring for oblique wave angles.
Radiation enhances directional sensitivity due to increased interference between incident and radiated components.
Recommendation: Spatial resolution must be selected such that directional effects are properly resolved. A minimum of 4–6 points per wavelength is required, particularly for oblique wave conditions.

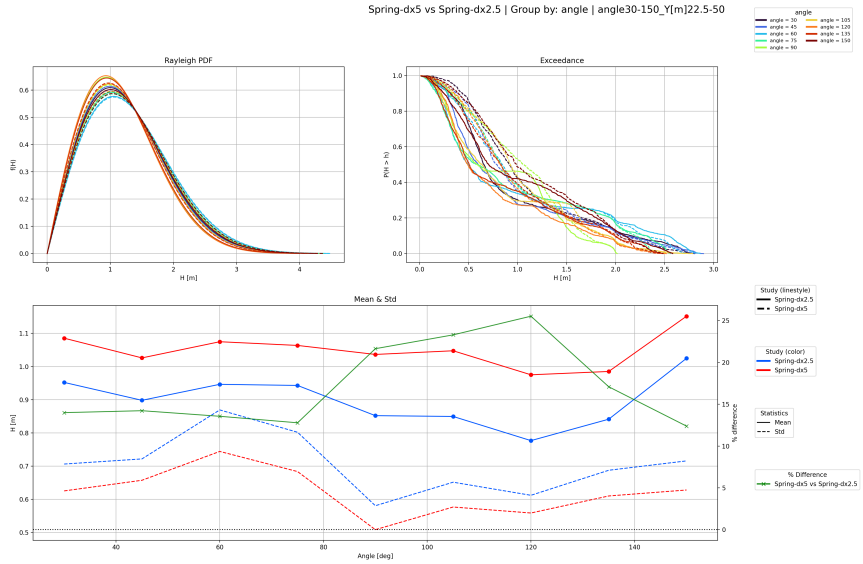


Figure 5.46: Effect of spatial resolution across wave directions.

5.2.3 Transverse Variation

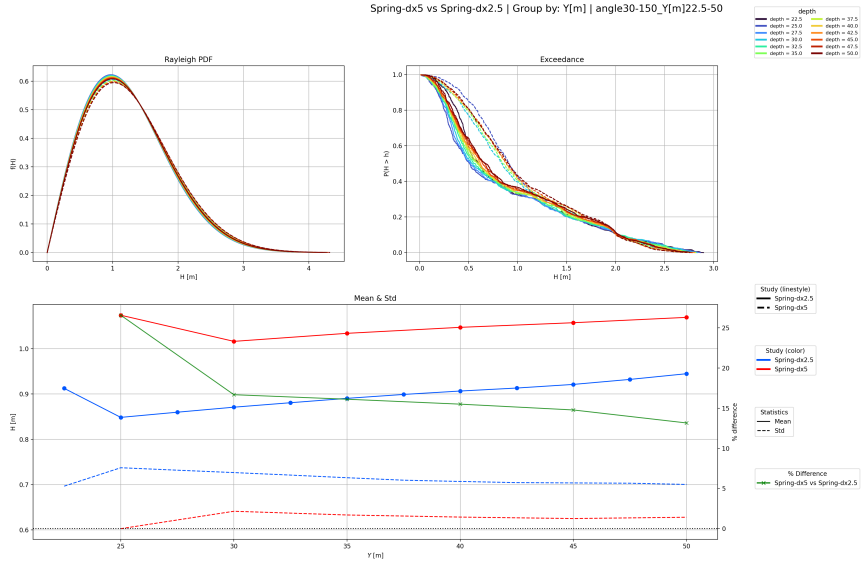


Figure 5.47: Wave statistics as a function of transverse position.

Figure 5.47 shows that both resolutions capture the same spatial trend along the transverse (y) direction. However, the coarse grid consistently overestimates the wave height by 10–20%.

The discrepancy is largest close to the vessel, where radiated waves dominate and spatial gradients are steep.

Recommendation: Grid refinement should be prioritised near the vessel and in regions of strong wave–structure interaction.

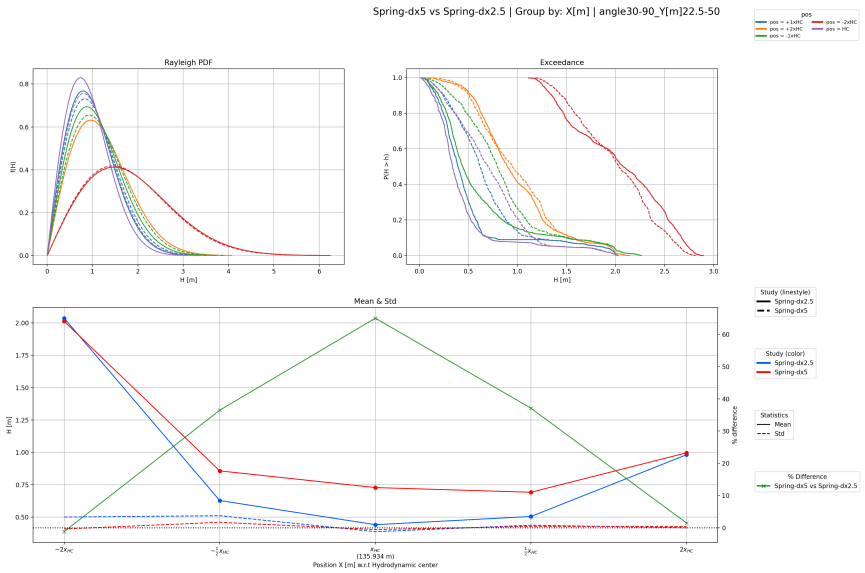


Figure 5.48: Spatial variation along x for angles $30°$–$90°$.

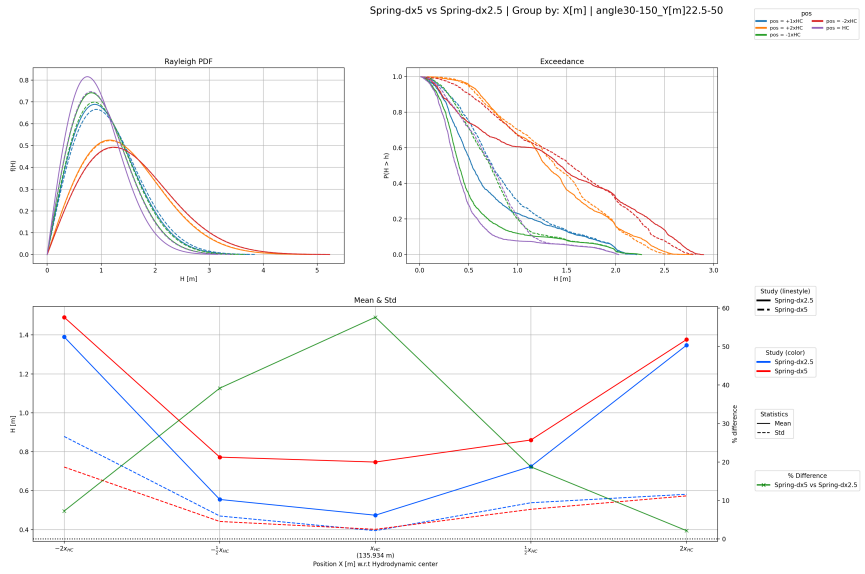


Figure 5.49: Spatial variation along x for angles $30°$–$150°$.

5.2.4 Longitudinal Variation

Figures 5.48 and 5.49 reveal strong spatial variability along the longitudinal direction. Near the hydrodynamic centre, differences between resolutions exceed 50%, indicating strong sensitivity to local gradients and radiated wave amplification.

Further away from the structure, discrepancies reduce but remain significant (~ 10–20%).

Recommendation:

- High spatial resolution is essential near the hydrodynamic centre.
- Local refinement strategies are recommended to reduce computational cost while preserving accuracy.

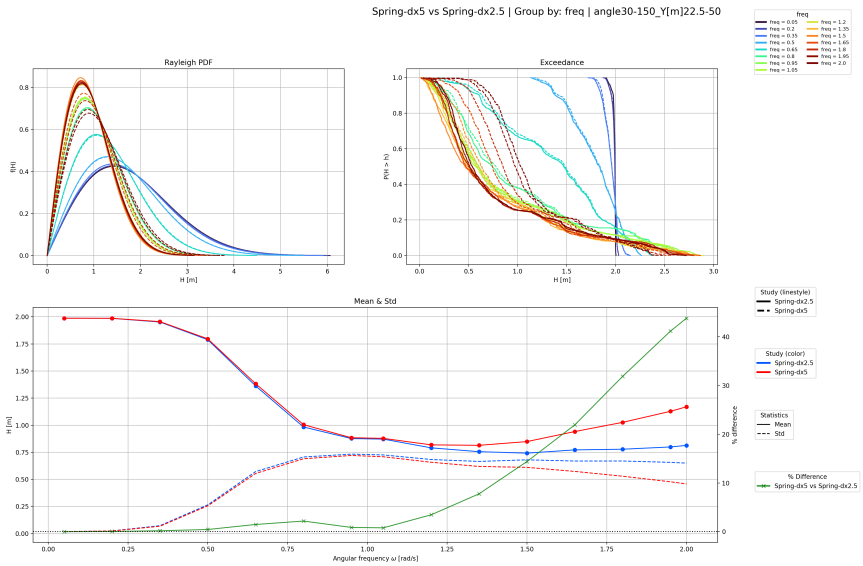


Figure 5.50: Effect of wave frequency on numerical accuracy.

5.2.5 Frequency Dependence

Figure 5.50 shows that spatial resolution errors strongly increase with frequency. At low frequencies, both grids provide similar results. However, at higher frequencies, discrepancies exceed 40% due to insufficient resolution of short wavelengths.
This effect is amplified by radiated waves, which inherently contain higher-frequency components.
Recommendation:

- Ensure at least 4–6 grid points per wavelength at the highest frequency of interest.
- Use adaptive or refined grids when high-frequency wave components are present.

5.3 Overall Discussion

The results clearly demonstrate that spatial discretization is the dominant source of numerical error in the presence of radiated waves. This is due to:

- The generation of short wavelengths by vessel motion,
- Increased coupling between fluid and structure,
- Strong spatial gradients near the vessel.

Temporal discretization plays a secondary role and becomes negligible once basic convergence is achieved.

5.4 Summary of Recommendations

- Use $\Delta t \approx T/100$ for temporal discretization,
- Ensure at least 4–6 points per wavelength in space,
- Prefer $\Delta x = 2.5\,\text{m}$ or finer for engineering accuracy,
- Apply local grid refinement near the vessel,

- Increase spatial resolution for high-frequency wave conditions,
- Consider adaptive meshing strategies to balance accuracy and cost.

5.5 Conclusion

The inclusion of radiated waves significantly increases sensitivity to spatial resolution, while temporal resolution remains a secondary factor. Coarse grids systematically overestimate wave heights and fail to capture localized amplification effects.
Accurate modelling of combined diffraction and radiation therefore requires sufficiently fine spatial discretization, particularly near the vessel and for high-frequency wave components.

5.6 Angular Interpolation Sensitivity Study

A dedicated sensitivity analysis was conducted to evaluate the accuracy and robustness of the angular interpolation procedure used to reconstruct the wave field for incident directions not explicitly available in the hydrodynamic database. The main objective is to quantify the error introduced by interpolation and to establish practical guidelines for angular discretization.
All figures are located in `Report/Figures/dwf/base/spring/comparedx5dx2.5/`.

5.6.1 Angular Dependence

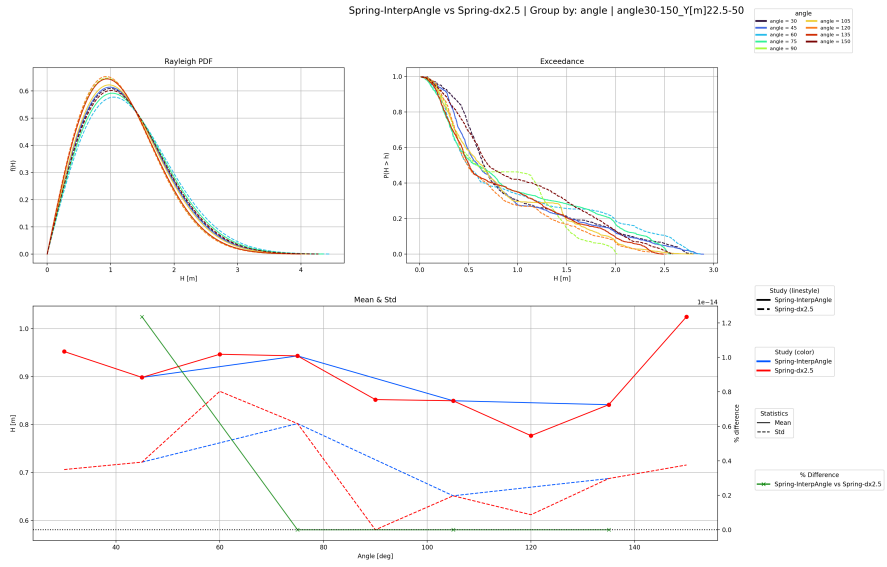


Figure 5.51: Comparison between interpolated and reference solutions across wave directions.

Figure 5.51 shows that the interpolated and reference solutions are in very good agreement across all incident angles. The Rayleigh probability density functions are nearly indistinguishable, with only minor differences in peak amplitude. The exceedance curves confirm this behaviour, with small deviations appearing only in the tail corresponding to extreme wave heights.
The mean wave height differences are negligible, remaining typically below 1–2%.
Interpretation: The directional dependence of the wave field remains sufficiently smooth, even in the presence of radiation, allowing accurate reconstruction through interpolation.
Recommendation: Angular discretization with steps up to 30° is adequate for most applications.

5.6.2 Transverse Dependence

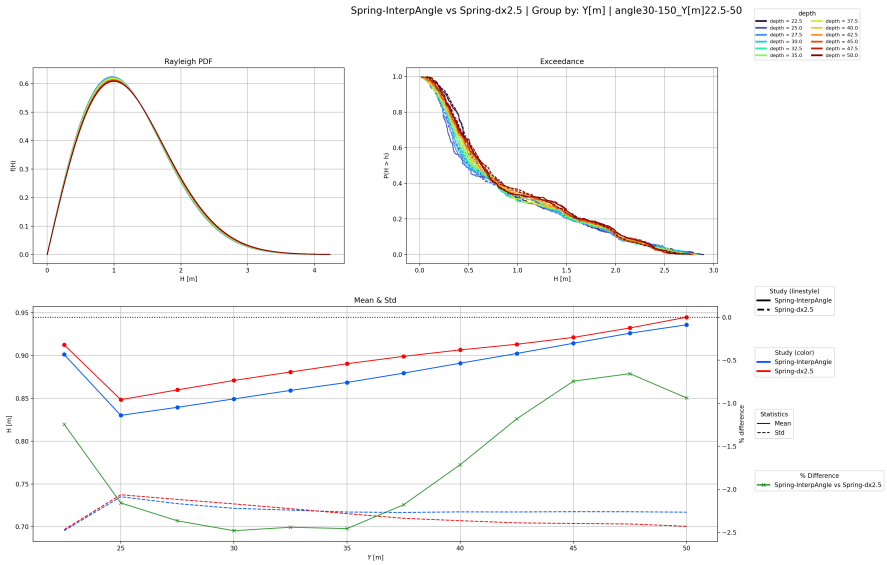


Figure 5.52: Angular interpolation sensitivity along the transverse direction.

Figure 5.52 indicates that interpolation accuracy is largely independent of transverse position. The statistical distributions remain consistent across all y locations, with differences in mean wave height below approximately 2%.
Slightly larger deviations are observed near the vessel, but these remain small compared to spatial discretization errors.
Recommendation: No additional angular refinement is required based on transverse position.

5.6.3 Longitudinal Dependence

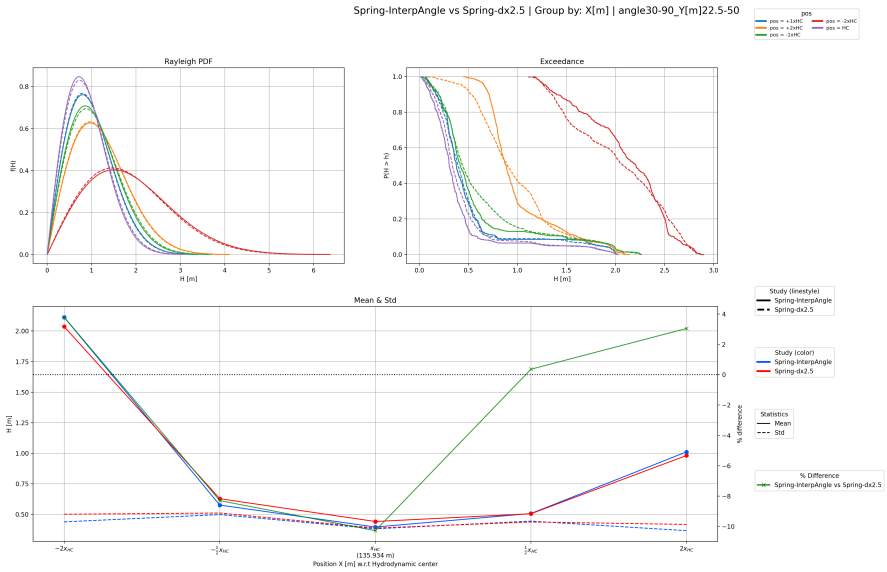


Figure 5.53: Angular interpolation sensitivity along x for angles $30°$–$90°$.

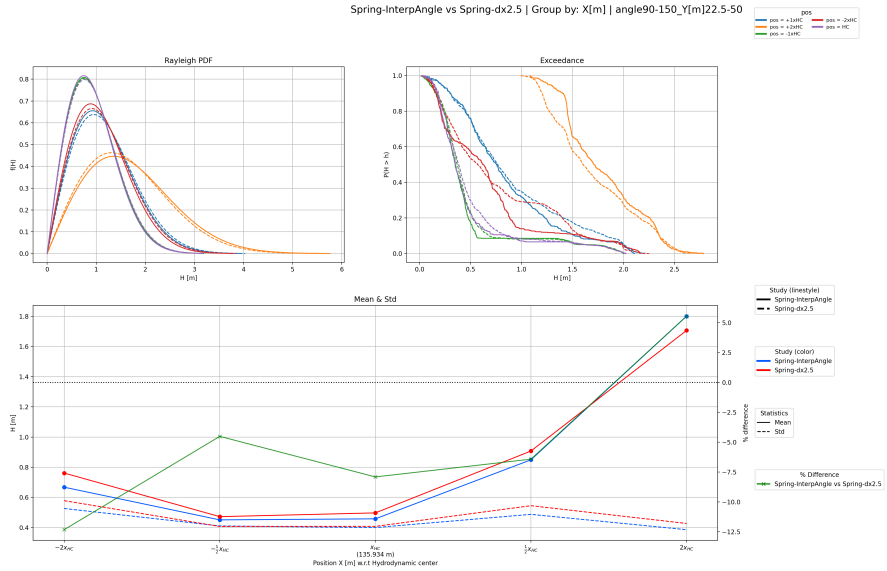


Figure 5.54: Angular interpolation sensitivity along x for angles $90°$–$150°$.

Figures 5.53 and 5.54 reveal a slightly stronger sensitivity to interpolation along the longitudinal (x) direction. While the interpolated and reference distributions remain qualitatively consistent, differences in mean wave height can locally reach values of approximately 5%–10%, particularly near the hydrodynamic centre. These discrepancies are associated with regions of strong spatial gradients and wave interference, where directional variations are more pronounced.

Interpretation: Angular interpolation errors are amplified in regions where the wave field varies rapidly in space, especially near the vessel.

Recommendation:

- Use finer angular resolution (e.g. $15°$) when accurate local predictions near the vessel are required,
- Coarser discretization remains acceptable in the far field.

5.6.4 Frequency Dependence

Figure 5.55 highlights the dependence of interpolation accuracy on wave frequency. At low frequencies, interpolated and reference results are nearly identical. As frequency increases, differences become more pronounced, with relative deviations fluctuating between 2% and 5%, and locally reaching higher values. These variations are linked to the increased directional variability associated with shorter wavelengths and radiated wave components.

Interpretation: Angular interpolation becomes less accurate at higher frequencies due to stronger directional gradients.

Recommendation:

- Increase angular resolution when high-frequency components dominate,
- Combine angular refinement with sufficient spatial resolution to properly resolve short waves.

5.6.5 Temporal Consistency

Figure 5.56 confirms that interpolation accuracy is independent of temporal resolution. Both interpolated and reference solutions converge simultaneously, and differences remain negligible across all tested time

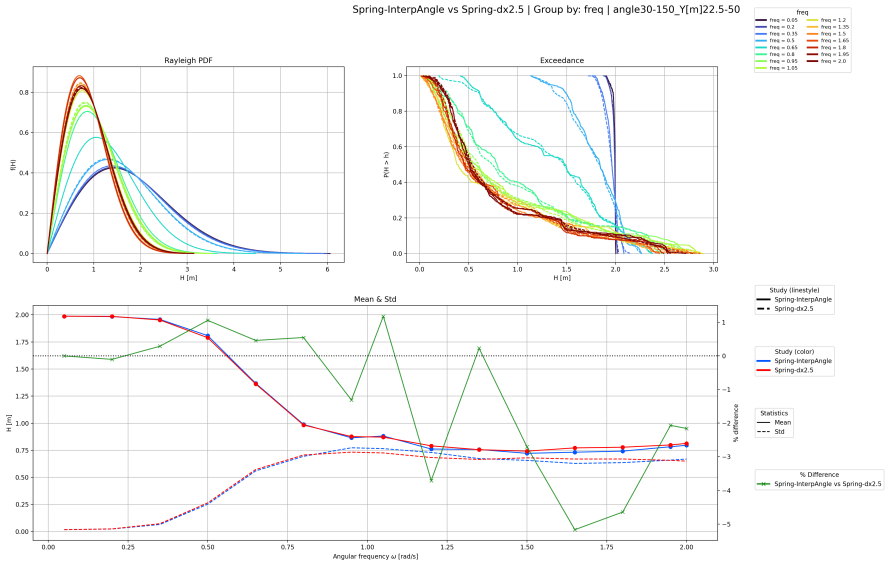


Figure 5.55: Angular interpolation sensitivity as a function of frequency.

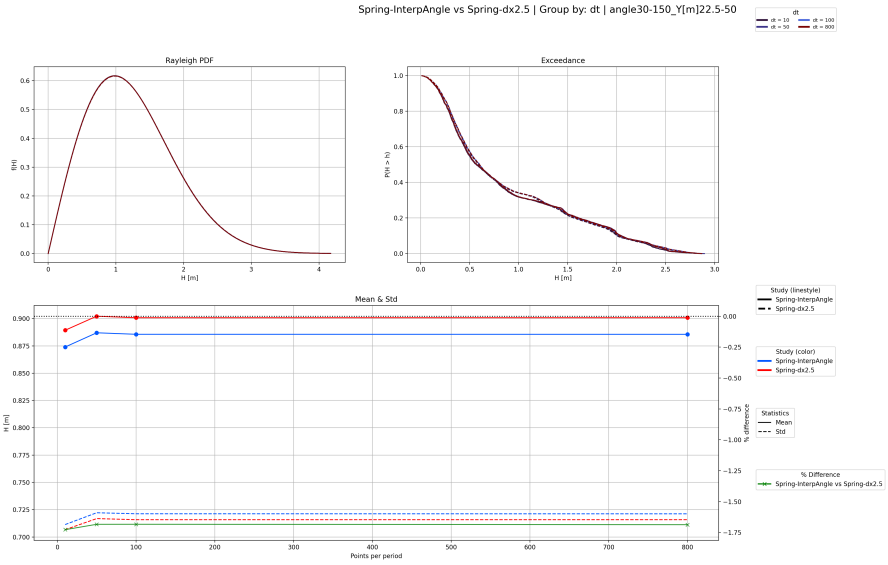


Figure 5.56: Effect of time step on interpolation accuracy.

steps.

Recommendation: Temporal discretization requirements are unaffected by angular interpolation.

5.6.6 Discussion and Engineering Implications

Overall, angular interpolation introduces relatively small errors compared to spatial discretization. While mean deviations are typically below 2%, localized discrepancies can reach up to 10% in regions characterised by strong spatial gradients or high-frequency content.

The robustness of the interpolation procedure is primarily due to the smooth variation of hydrodynamic response with angle. However, this assumption becomes weaker for:

- High-frequency wave components,
- Regions close to the vessel,

- Strongly directional wave conditions.

5.6.7 Guidelines for Angular Discretization

Based on the present analysis, the following practical recommendations are proposed:

- Use angular steps of 30° for general engineering applications,
- Reduce to 15° when high-frequency waves or strong local gradients are present,
- Prioritise spatial resolution over angular refinement, as it remains the dominant error source,
- Apply finer angular discretization near the vessel if local accuracy is critical.

5.6.8 Conclusion

Angular interpolation is shown to be robust and accurate for most practical applications, with relatively small impact on global wave statistics. However, its accuracy decreases slightly in regions of strong spatial variability and for high-frequency wave components.
Compared to spatial discretization, interpolation remains a secondary source of error, but targeted refinement may be required for high-fidelity local analyses.

5.7 Interpolation Spatial Step Study

In this section, the influence of spatial interpolation on the evaluation of the wave field is analysed. The wave probe locations are shifted by $1\,\mathrm{m}$ relative to the computational grid, requiring the Diffraction Wave Field (DWF) model to interpolate the free-surface elevation at off-grid positions. All simulations are performed using a spatial resolution of $\Delta x = 2.5\,\mathrm{m}$, which was previously identified as providing converged results.
All figures are located in `Report/Figures/dwf/base/spring/comparedx5dx2.5/`.

5.7.1 Transverse Dependence

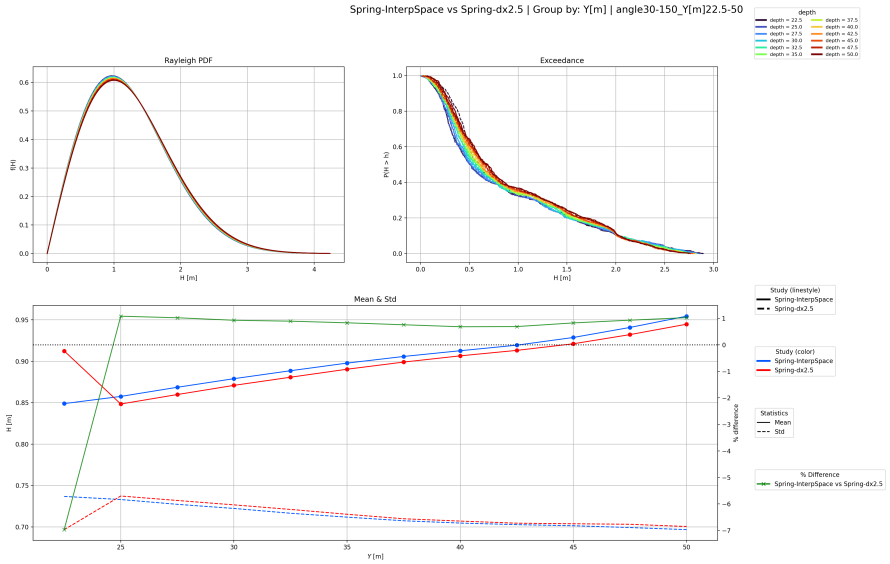


Figure 5.57: Effect of spatial interpolation as a function of transverse position.

Figure 5.57 shows that spatial interpolation introduces only minor deviations along the transverse (y) direction. The Rayleigh probability density functions and exceedance curves are nearly indistinguishable between interpolated and reference probe locations.

Differences in mean wave height remain typically below 1–2%, with slightly larger deviations occurring at smaller y values. These correspond to regions closer to the vessel, where spatial gradients are stronger due to radiated wave contributions.

Interpretation: Spatial interpolation accuracy is only weakly affected by transverse position and remains robust even in regions influenced by vessel-induced wave components.

Recommendation: Standard interpolation schemes are sufficient along the transverse direction when an adequately refined grid is used.

5.7.2 Longitudinal Dependence

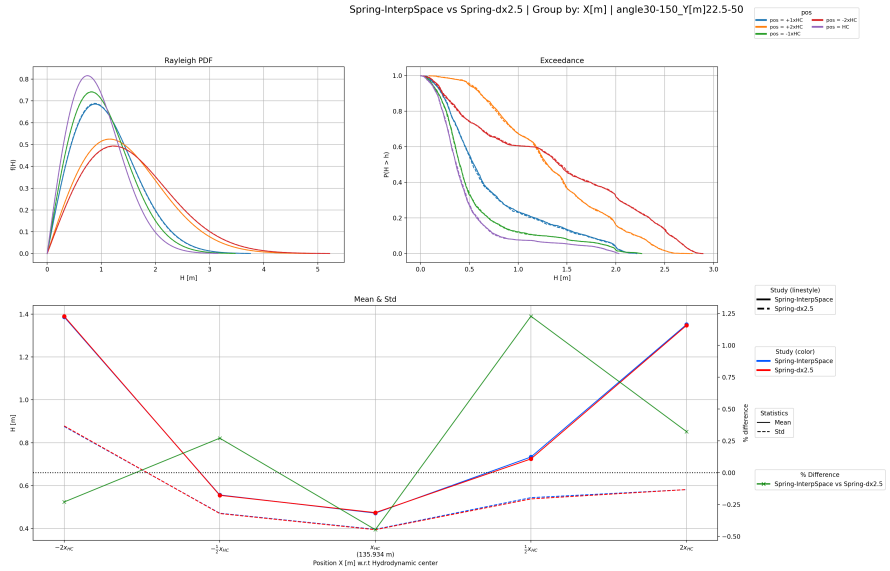


Figure 5.58: Effect of spatial interpolation along the longitudinal (x) direction.

Figure 5.58 highlights the spatial interpolation behaviour along the longitudinal direction. Compared to the transverse case, slightly larger variations are observed, reflecting stronger gradients and wave interference effects along x.

Nevertheless, the overall agreement remains very good. Differences in mean wave height are generally within 1–3%, with localized variations depending on the probe position relative to the hydrodynamic centre.

Interpretation: Interpolation errors are slightly amplified along x due to stronger spatial variability, but remain small compared to spatial discretization errors.

Recommendation:

- Off-grid evaluation along x is reliable,
- Additional care may be required near the hydrodynamic centre if very high accuracy is needed.

5.7.3 Angular Dependence

Figure 5.59 shows that spatial interpolation introduces minimal directional bias. The statistical distributions are preserved across all incident wave angles, with only slight variations in peak values.

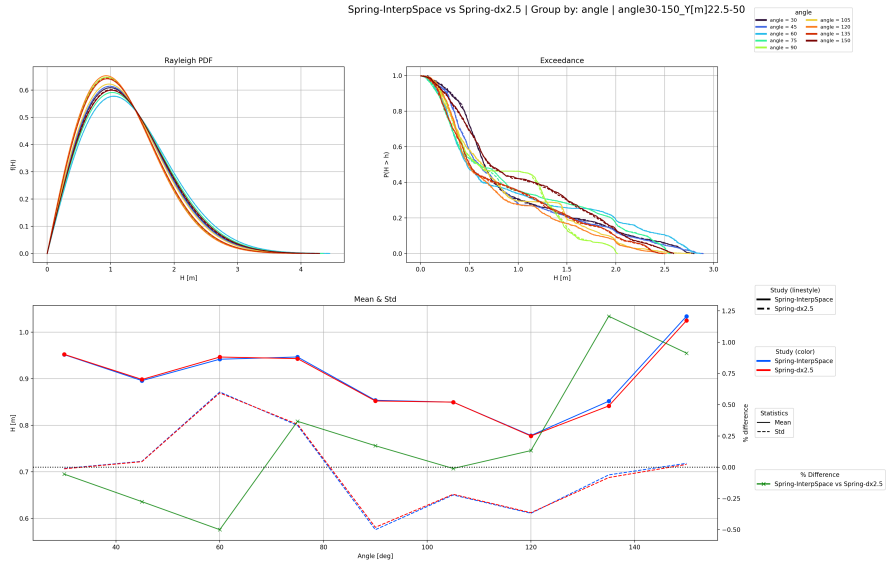


Figure 5.59: Spatial interpolation sensitivity across wave directions.

The differences in mean wave height are typically below 2%, although small fluctuations appear for oblique wave directions where interference patterns are more complex.
Interpretation: Spatial interpolation is robust with respect to incident wave direction, with limited sensitivity to directional effects.
Recommendation: No additional angular refinement is required to improve spatial interpolation accuracy.

5.7.4 Temporal Consistency

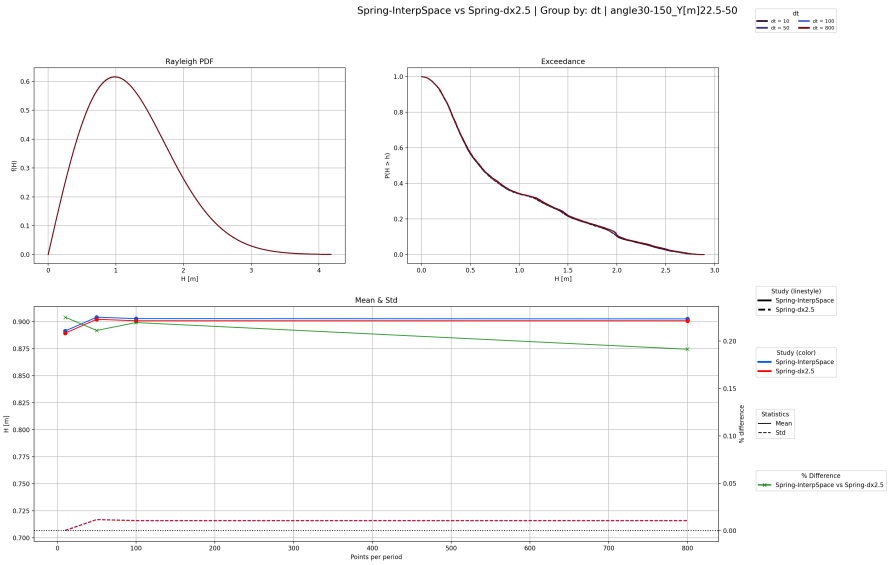


Figure 5.60: Effect of time step on spatial interpolation accuracy.

Figure 5.60 confirms that spatial interpolation errors are insensitive to temporal discretization. Both interpolated and reference solutions converge simultaneously with increasing temporal resolution.
Interpretation: Spatial interpolation does not introduce coupling with temporal discretization errors.
Recommendation: Previously established temporal resolution criteria ($\Delta t \approx T/100$) remain valid.

5.7.5 Frequency Dependence

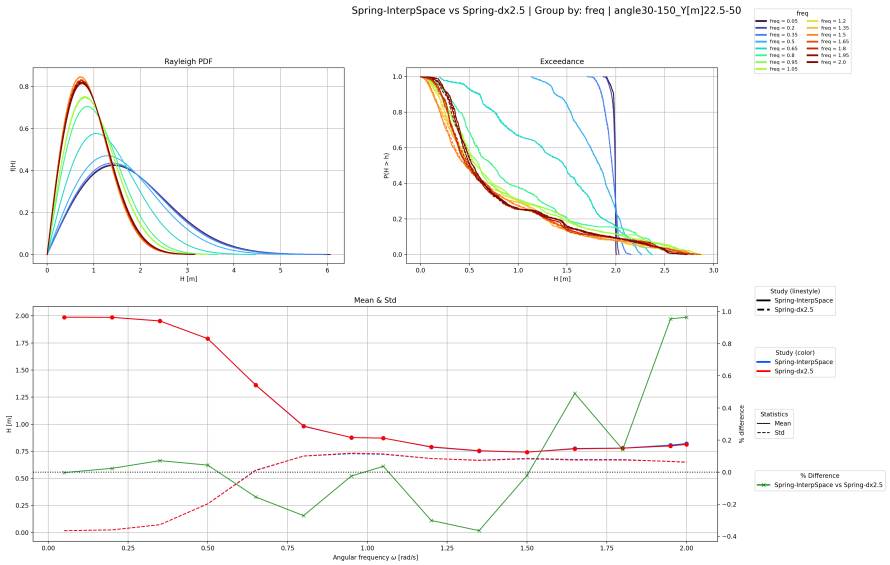


Figure 5.61: Spatial interpolation sensitivity as a function of wave frequency.

Figure 5.61 illustrates the effect of spatial interpolation across frequencies. At low and intermediate frequencies, the agreement between interpolated and reference results is nearly exact. At higher frequencies, slight deviations emerge, reflecting the reduced spatial resolution relative to wavelength.
The observed differences remain small, typically below 2–3%, even at the highest frequencies considered.
Interpretation: Interpolation accuracy decreases mildly for short waves, but remains within acceptable engineering tolerances.
Recommendation:

- Ensure sufficient spatial resolution for high-frequency components,
- No modification of the interpolation scheme is required.

5.7.6 Discussion and Practical Implications

The results demonstrate that spatial interpolation is a secondary source of error when a sufficiently fine computational grid is employed. Across all analysed dimensions (space, direction, frequency, and time), interpolation-induced differences remain small, typically within 1–3%.
Slightly higher sensitivity is observed:

- Along the longitudinal direction (x),
- Near the vessel, where gradients are strongest,
- At higher frequencies, where wavelengths are shorter.

However, these effects remain limited and do not significantly affect overall wave statistics.

5.7.7 Guidelines for Spatial Interpolation

Based on the present analysis, the following recommendations can be formulated:

- Spatial interpolation is reliable for off-grid probe evaluation when $\Delta x = 2.5\,\mathrm{m}$ or finer,
- Interpolation errors are typically below 3%, significantly smaller than discretization errors,
- No additional refinement in angle or time is required to improve interpolation accuracy,
- Slightly finer grids may be beneficial for high-frequency or strongly localised phenomena,
- Priority should remain on spatial discretization rather than interpolation method.

5.7.8 Conclusion

Spatial interpolation is shown to be accurate and robust across all tested conditions. Even in the presence of radiated waves and strong local gradients, the introduced errors remain small and well within acceptable limits.
Compared to spatial discretization, interpolation has a negligible influence on the solution accuracy. Consequently, computational effort should be focused on achieving sufficient grid resolution rather than refining interpolation strategies.

5.8 Comparison with Variational Boussinesq Model (VBM)

A comparison is performed between the Diffraction Wave Field (DWF) model and the Variational Boussinesq Model (VBM) to assess the accuracy of the reconstructed wave field when both diffraction and radiation effects are included. The analysis focuses on statistical wave properties, including Rayleigh probability density functions, exceedance curves, and bulk metrics such as mean and standard deviation.
All figures are located in `Report/Figures/dwf/base/spring/comparedx5dx2.5/`.

5.8.1 Angular Dependence

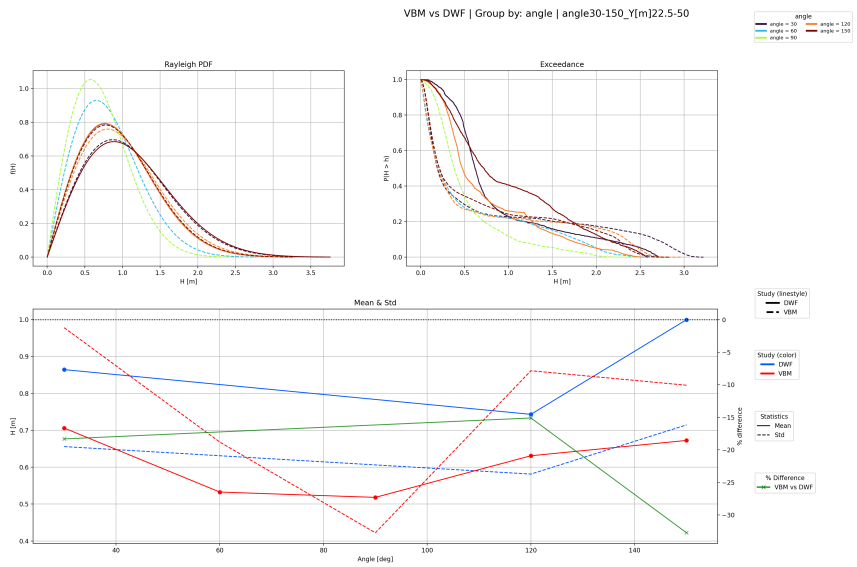


Figure 5.62: Comparison between DWF and VBM across wave directions.

Figure 5.62 shows that both models capture similar trends in wave statistics across incident directions. The shape of the Rayleigh distributions is consistent, indicating that both approaches reproduce the underlying stochastic structure of the wave field.

However, a systematic deviation is observed in wave amplitude. VBM consistently predicts lower wave heights compared to DWF. The relative differences range from approximately 10% to 30%, with the largest discrepancies occurring for oblique wave directions where radiation–diffraction coupling is strongest.

Interpretation: Directional behaviour is well reproduced, but VBM underestimates wave amplitude, particularly for directions associated with strong wave–structure interaction.

5.8.2 Transverse Dependence

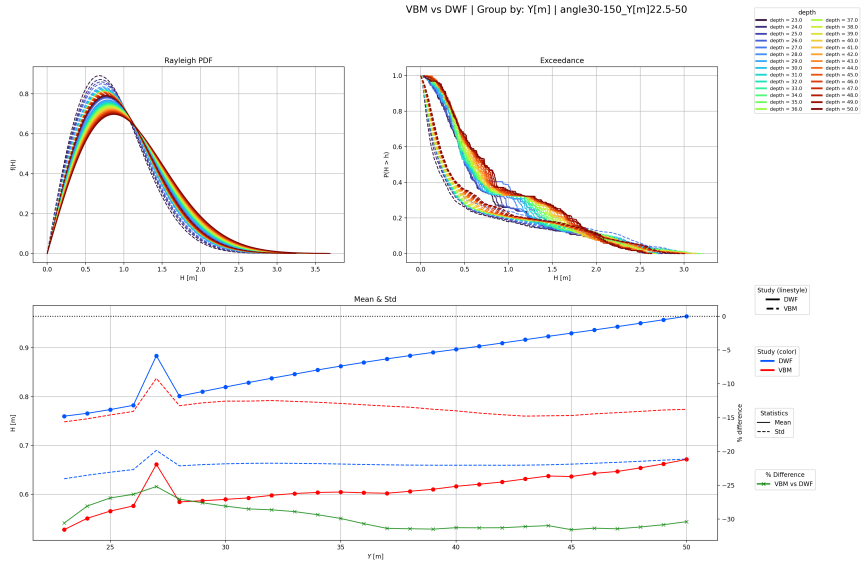


Figure 5.63: Comparison between DWF and VBM along the transverse direction.

Figure 5.63 presents the variation of wave statistics along the transverse (y) direction. Both models exhibit a consistent spatial trend, with wave height increasing gradually away from the vessel.

Despite this agreement, VBM systematically underestimates the wave height across all transverse positions. The difference is relatively uniform, typically in the range of 20–30%. This suggests that the discrepancy is not strongly dependent on transverse location, but instead reflects a global modelling difference.

Interpretation: Both models capture the same spatial evolution, but VBM predicts consistently lower amplitudes.

5.8.3 Longitudinal Dependence

Figures 5.64–5.66 highlight the longitudinal variation of the wave field. This direction exhibits the largest discrepancies between the two models.

Near the hydrodynamic centre ($x \approx 0$), relative differences remain moderate ($\sim$ 20–40%). However, away from the centre, particularly for large $|x|$, the deviations increase significantly and can exceed 50%–60% in some cases.

The exceedance curves show that VBM strongly underpredicts extreme wave events compared to DWF, especially in regions influenced by radiated wave interference patterns.

Interpretation: The longitudinal direction is the most sensitive to modelling differences. VBM fails to accurately capture localized amplification and interference effects generated by vessel motion.

Key insight: Discrepancies are largest precisely where radiated waves and spatial gradients are strongest.

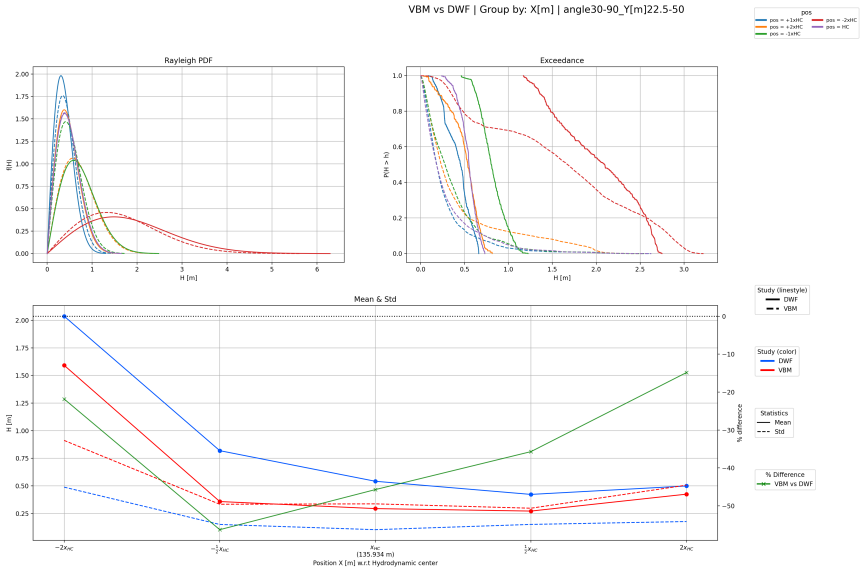


Figure 5.64: Comparison between DWF and VBM along x for angles $30°$–$90°$.

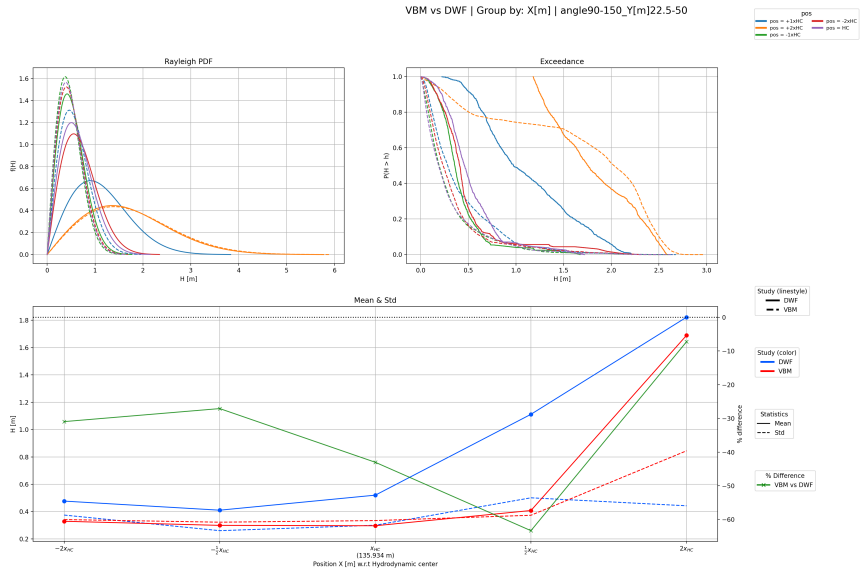


Figure 5.65: Comparison between DWF and VBM along x for angles $90°$–$150°$.

5.8.4 Frequency Dependence

Figure 5.67 demonstrates that discrepancies between DWF and VBM increase significantly with frequency. At low frequencies, differences are moderate (~ 10–20%). As frequency increases, deviations grow steadily, reaching 30–40% and exceeding this range at the highest frequencies.

The exceedance curves indicate that VBM predicts significantly lower extreme wave heights, especially for short waves.

Interpretation: VBM is less effective at resolving short-wave dynamics and radiated wave components, which dominate at higher frequencies.

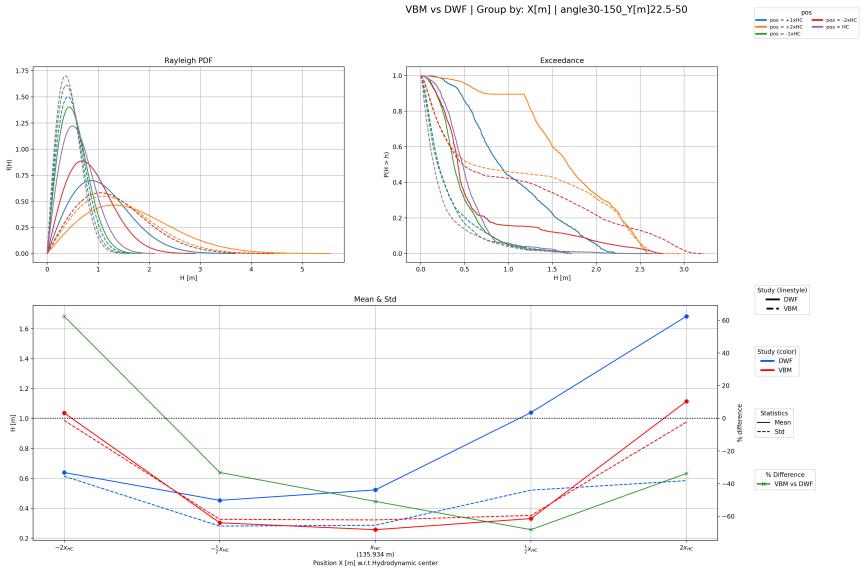


Figure 5.66: Comparison between DWF and VBM along x for the full angular range.

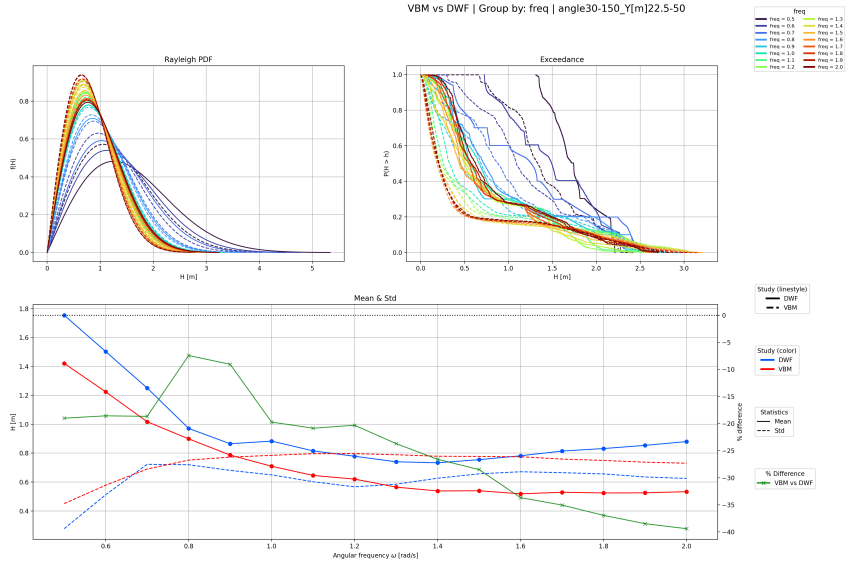


Figure 5.67: Comparison between DWF and VBM as a function of wave frequency.

5.8.5 Discussion

The comparison shows that DWF and VBM provide consistent qualitative predictions of wave behaviour but differ quantitatively.
Key findings include:

- DWF consistently predicts higher wave heights than VBM,
- Differences increase with frequency and can exceed 40%,
- Longitudinal variations show the largest discrepancies (up to 60%),
- Transverse variations show more uniform differences (20–30%),
- Extreme wave statistics are significantly underestimated by VBM.

These discrepancies can be attributed to the different physical representations of wave processes. The DWF model explicitly includes diffraction and radiation effects, while the VBM formulation introduces approximations that tend to smooth out short-wavelength components and reduce localized amplification.

5.8.6 Implications for Modelling

- VBM is suitable for capturing large-scale trends and qualitative behaviour,
- DWF is more accurate for predicting wave amplitudes and extremes,
- The difference is most critical in the near field and for high-frequency waves,
- Applications requiring detailed local wave fields (e.g. loads, mooring response) should favour DWF.

5.8.7 Conclusion

The comparison confirms that both models reproduce the same physical mechanisms governing diffraction and radiation. However, DWF provides a more accurate representation of wave amplitudes, particularly in regions of strong spatial gradients and high-frequency conditions.
VBM systematically underestimates wave heights, with discrepancies ranging from 10–30% in general conditions to more than 50% in localized areas. While VBM remains suitable for global trend analysis, accurate prediction of wave extremes and local amplification requires higher-fidelity approaches such as DWF.

5.9 Global Conclusions and Practical Recommendations

This section has investigated the modelling of combined diffraction and radiation effects, with a particular focus on numerical sensitivity, interpolation strategies, and comparison against the Variational Boussinesq Model (VBM). The results consistently highlight the existence of a clear hierarchy in the sources of numerical and modelling error.

5.9.1 Overall Conclusions

The main conclusions can be summarised as follows:

- **Spatial discretization is the dominant source of numerical error.** The inclusion of radiated waves introduces short wavelengths and strong spatial gradients, which require fine spatial resolution. Insufficient resolution leads to systematic overestimation of wave heights, with differences ranging from 10–25% in general conditions and exceeding 50% locally, particularly along the longitudinal direction and near the vessel.
- **Temporal discretization plays a secondary role.** Once a moderate temporal resolution is achieved ($\Delta t \leq T/100$), the solution is effectively converged and temporal errors remain below 5%.
- **Interpolation effects are limited.** Both angular and spatial interpolation introduce relatively small errors. Mean discrepancies remain typically below 2–3%, with local variations up to 5–10% in regions of strong gradients or high-frequency content. These errors are negligible compared to discretization effects.
- **Model choice significantly affects amplitude prediction.** The comparison with VBM shows that, although global trends are consistently captured, VBM systematically underestimates wave heights.

Differences range from 10–30% in general conditions and can exceed 50% in regions dominated by radiation and interference phenomena.

- **High-frequency and near-field regions are the most critical.** The largest discrepancies occur for high-frequency waves and in the vicinity of the vessel, where radiated waves generate localized amplification and strong spatial variability.

5.9.2 Recommended Modelling Strategy

Based on the combined results of all analyses, the following practical guidelines are proposed:

- **Temporal resolution:** Use $\Delta t \approx T/100$. Further refinement is generally unnecessary.
- **Spatial resolution:** Ensure at least 4–6 grid points per wavelength, with $\Delta x \leq 2.5\,\mathrm{m}$ for the present configuration. Apply local refinement near the hydrodynamic centre and along the longitudinal direction.
- **Angular discretization:** Use angular steps of 30° for standard applications. Reduce to 15° for high-frequency waves or when accurate near-field predictions are required.
- **Interpolation practice:** Standard interpolation schemes are sufficient. No additional refinement is needed, provided the underlying spatial grid is adequately resolved.
- **Model selection:**
  - Use DWF for accurate prediction of wave amplitudes, local amplification, and extreme statistics,
  - Use VBM for qualitative analysis and large-scale trends when computational efficiency is prioritised.

5.9.3 Final Remarks

The combined diffraction–radiation problem is primarily controlled by the ability of the numerical model to resolve short-wave dynamics and spatial gradients. While interpolation and temporal discretization introduce only minor errors, spatial resolution and modelling approach play a critical role in determining accuracy.
For applications involving wave loads, vessel response, or extreme wave statistics, high-resolution spatial discretization and high-fidelity models such as DWF are essential. Conversely, simplified models such as VBM remain valuable for preliminary assessments, provided their limitations in predicting wave amplitudes are properly accounted for.

6 Conclusion

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REFERENCES

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TABLES

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Validation Diffraction Node in XMF

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FIGURES

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